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A Minimax Rescheduling Model for Smoothing Time-Dependent Truck Demand in B2B Delivery Scheduling

Master's Thesis

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List of Abbreviations

B2B	Business-to-business
MILP	Mixed-integer linear programming
OR	Operations research
RMC	Ready-mixed concrete
VRP	Vehicle routing problem
VRPTW	Vehicle routing problem with time windows

List of Symbols

Symbol	Description
I	Set of customers scheduled on the considered day.
T	Set of discrete time slots in the daily planning horizon.
K_i	Set of feasible shift options for customer i.
i, t, k, u	Customer, time-slot, shift-option, and relative profile-position indices.
a_i	Original start slot of customer i.
m_i	Number of active slots in the demand profile of customer i.
p_{iu}	Demand of customer i at relative profile position u.
q_{itk}	Contribution of customer i to slot t if shift k is selected.
B	Maximum number of customers allowed to be moved in the deterministic model.
Δ_i	Maximum absolute shift allowed for customer i.
c_i, h_i	Fixed and shift-magnitude disruption penalty parameters.
x_{ik}	Binary variable equal to 1 if customer i is assigned shift k.
y_i	Binary indicator equal to 1 if customer i is moved.
L_t	Aggregate truck demand in slot t.
Z	Maximum aggregate truck demand over the day.
s_i	Absolute shift size assigned to customer i.
$C^{\text{own}}, C^{\text{long}}$	Own-fleet capacity and long-term rented-capacity threshold.

Symbol	Description
$c^{\text{long}}, c^{\text{spot}}$	Cost coefficients for long-term rented and spot-market capacity.
G_t, H_t	Long-term rented and spot-market capacity variables in the optional capacity-tier extension.
F, R	Fixed and recovery-eligible customer sets in the recovery stage.
B^R	Maximum number of additional customers allowed to be changed in recovery.
$\bar{k}_i$	Fixed recovery-stage shift for customer $i \in F$.
x^R_{ik}, y^R_i, Z^R	Recovery-stage shift, change-indicator, and peak variables.

1 Introduction

Many delivery systems face a recurring mismatch between customer-preferred time windows and the operational capacity available to serve them. In attended and time-constrained delivery settings, narrow delivery windows improve service convenience but reduce the operational freedom available for routing, scheduling, and capacity use (Campbell and Savelsbergh, 2005, 2006; Agatz et al., 2011). Flexible delivery-time concepts and demand-management approaches therefore show that temporal flexibility can have substantial operational value, especially when delivery choices interact with fulfilment costs (Asdemir et al., 2009; Yang et al., 2016; Strauss et al., 2021).

This thesis studies a related but more focused business-to-business (B2B) setting using an empirically motivated generated dataset of customer-level delivery requests over several planning days. Each request is represented by a start time and a time-dependent truck demand profile over 15-minute time slots. The profile describes how much capacity the customer contributes to each active time slot. The central operational issue is therefore not that total demand is too high in aggregate. Rather, the problem is that the temporal placement of demand creates peaks in simultaneous truck requirements. These peaks may increase reliance on external capacity, create idle time in less popular periods, or make the daily plan more difficult to operate.

The problem is best understood as a time-dependent capacity-smoothing problem. Resource-leveling research studies how activities can be shifted within feasible ranges to smooth resource utilization over time, while minimax allocation provides a natural objective when the planner wants to reduce the largest resource requirement in a system (Klein et al., 1993; Rieck et al., 2012). In addition, the rescheduling literature emphasizes that revising an existing plan is not the same as planning from scratch: schedule improvements should be balanced against deviations from the original plan (Vieira et al., 2003; Wu et al.,

1993; Rangsaritratsamee et al., 2004). These two ideas define the analytical core of the thesis. Existing customer demand profiles are repositioned within limited flexibility windows in order to reduce the maximum simultaneous truck requirement, while the number and magnitude of schedule changes are explicitly controlled.

The available data contain temporal demand profiles, but they do not contain customer locations, travel times, depots, driver rosters, route sequences, or truck-specific assignments. Consequently, the thesis focuses on the part of the planning problem that can be modelled rigorously with the available information: the rescheduling of customer demand profiles across a discrete daily timeline. This abstraction is also consistent with timed-delivery applications such as ready-mixed concrete delivery, where the timing of truck requirements is operationally important, although full dispatching models usually include richer routing and plant-assignment details (Yan et al., 2008; Kinable et al., 2014).

The primary research objective is to develop and test a deterministic minimax rescheduling model for reducing peak simultaneous truck demand in a B2B delivery setting. The model should provide a transparent decision-support structure for identifying which customer schedules should be shifted and for quantifying the trade-off between peak reduction and schedule disruption. The main research question is formulated as follows:

> How can customer delivery schedules be rescheduled to minimize peak truck demand under limited customer flexibility and minimal schedule disruption?

This main question is decomposed into four sub-questions:

1. Which customer demand profiles should be shifted to reduce the daily peak most effectively?

2. How much peak reduction can be achieved when each customer has only limited rescheduling flexibility?

3. What is the trade-off between peak reduction and the number or magnitude of customer schedule changes?

4. How much additional smoothing can be restored by a limited recoverable rescheduling step if part of the first-stage plan cannot be implemented?

The first three sub-questions define the deterministic core of the thesis. The fourth sub-question is a focused extension around the same model. It does not turn the thesis into a broad uncertainty or customer-behaviour study; it only tests whether the deterministic minimax formulation can also support a limited second-stage recovery action.

The thesis makes a focused contribution at the intersection of rescheduling, resource smoothing, and timed delivery operations. It does not attempt to solve the complete logistics problem of B2B deliveries. In particular, the model does not construct routes, assign individual vehicles, or optimize travel paths. Instead, it isolates the temporal capacity-smoothing component of the delivery planning problem. This scope has two advantages. First, it matches the structure of the available data. The data directly support the construction of aggregate load curves over time, but they do not support a full vehicle routing formulation. Second, it creates a clean modelling contribution. The thesis studies a profile-placement problem: fixed customer demand shapes are repositioned in a daily planning horizon in order to reduce the highest point of the aggregate load curve.

The expected contribution is threefold. First, the thesis formulates a deterministic minimax model that assigns exactly one feasible final position to each customer profile. Second, it extends the base formulation with disruption-aware components, such as a cap on the number of changed customers and lexico-

graphic secondary objectives. Third, it evaluates a recoverable rescheduling extension in which a limited second-stage adjustment can be made if part of the initial plan cannot be implemented. The intended output is not only an optimal load curve, but also managerial insight into how much rescheduling effort is required to obtain a given level of peak reduction.

The remainder of the thesis is structured as follows. Section 2 reviews the relevant literature streams and positions the thesis relative to rescheduling, resource leveling, flexible delivery time windows, ready-mixed concrete delivery, and robust scheduling. Section 3 describes the planning problem, the data structure, the rescheduling logic, an illustrative profile-placement example, and the assumptions that delimit the scope of the model. Section 4 develops the mathematical formulation, including the deterministic minimax model, disruption-aware variants, and the recoverable rescheduling extension. Section 5 presents the numerical study, including deterministic peak-smoothing results, disruption-budget experiments, day-level variation, and recoverable rescheduling. Section 6 derives managerial insights, and Section 7 concludes the thesis.

2 Literature Review

This chapter positions the thesis at the intersection of rescheduling, resource smoothing, and timed delivery operations. The central problem is narrower than a full vehicle-routing or dispatching problem: already scheduled customer demand profiles are shifted within limited flexibility windows in order to reduce the maximum simultaneous truck requirement. The relevant literature is therefore organized around four connected ideas that are discussed in a more compact structure. First, rescheduling explains why revising an existing plan is a distinct planning problem. Second, stability-oriented scheduling clarifies why disruption must be controlled. Third, resource leveling and minimax allocation provide the closest mathematical analogy for smoothing a time-dependent load

profile. Fourth, timed-delivery and ready-mixed-concrete studies motivate the operational importance of truck-capacity peaks. Recoverable rescheduling is then used only as a focused extension around the same core model.

2.1 Rescheduling and Disruption Control

The starting point for the thesis is the rescheduling literature. Rescheduling refers to the revision of an existing schedule when the original plan no longer performs well, becomes infeasible, or can be improved after new information becomes available. In production environments, the initial schedule provides coordination and stability, but real systems are dynamic and often require schedule revisions. Vieira et al. (2003) offer one of the most important conceptual frameworks for rescheduling, distinguishing rescheduling strategies, policies, and methods. Their framework is useful for the present thesis because the problem is not a greenfield scheduling problem. The original customer schedules already exist, and the planner considers controlled changes to this existing plan.

Earlier work on dynamic scheduling makes a related distinction between predictive and reactive scheduling. Predictive scheduling generates a plan before execution, whereas reactive scheduling responds to disruptions or new information during execution (Suresh and Chaudhuri, 1993). Ouelhadj and Petrovic (2009) further emphasize that static scheduling methods are often insufficient for dynamic real-world environments, where schedules must be reconsidered as new information appears. Although the present thesis is not primarily about real-time shop-floor disruptions, the same logic applies: an initial delivery schedule can be operationally undesirable because demand is concentrated in a few time intervals. Rescheduling is therefore used as a structured intervention to improve the temporal allocation of demand.

Related scheduling research broadens this view to stochastic, robust, and

predictive-reactive settings. Classical sequencing and stochastic scheduling studies establish the broader scheduling foundation on which rescheduling models build (Graham et al., 1979; Pinedo, 1983; Aytug et al., 2005). Robust and uncertainty-oriented scheduling research then shows why schedules should be designed or revised with uncertain processing and project conditions in mind (Daniels and Kouvelis, 1995; Herroelen and Leus, 2005). Further work on local, event-driven, and predictive-reactive rescheduling emphasizes that schedule revision is a distinct operational decision rather than a simple re-run of the original scheduling model (Vieira et al., 2000; Kuster et al., 2010; Tighazoui et al., 2021).

The multi-objective nature of rescheduling is especially important. Rescheduling studies emphasize that schedule revisions should consider both operational efficiency and stability, rather than treating the revised schedule as a purely performance-oriented solution. This point maps directly onto the thesis. The planner wants to reduce peak truck demand, but a plan that moves too many customers or imposes large shifts may be operationally unattractive. The literature therefore supports modelling the problem not only as a peak-minimization problem, but also as a schedule-change problem. In practical terms, this means that the model should restrict the number of changed customers, penalize shift magnitude, or apply a lexicographic objective in which peak reduction comes first and disruption minimization comes second.

A central lesson from the rescheduling literature is that a new schedule should not be evaluated only by its operational efficiency. It should also be evaluated by how much it deviates from the original schedule. Wu et al. (1993) formulate one-machine rescheduling as a bicriteria problem involving schedule efficiency and schedule stability. They measure stability through deviations from the original schedule, including start-time deviations and sequence differences. Their results are important for this thesis because they show that the cost of

rescheduling is not merely psychological or managerial; it can be formalized as a modelling criterion.

Rangsaritratsamee et al. (2004) develop the same idea in a dynamic job-shop context. They argue that periodic rescheduling may improve classical efficiency measures, but excessive instability can make a technically efficient plan impractical on the shop floor. This reasoning is directly transferable to B2B delivery scheduling. A solution that reduces the peak by shifting many customer profiles across the day may be mathematically attractive, but it may create too many customer contacts, negotiation efforts, or service-quality risks. Therefore, the thesis treats rescheduling disruption as a first-class modelling concern.

This trade-off has two modelling implications. First, the model should include a hard or soft control on the number of changed customers. A hard cap is easy to communicate: the planner allows at most B customer schedules to change. A soft penalty is more flexible: the objective penalizes every changed customer. Second, the model should distinguish small changes from large changes. A 15-minute shift is usually less disruptive than a two-hour shift. The literature on schedule stability supports this because stability is often measured by deviation from the original schedule, not only by whether an activity changed at all (Wu et al., 1993).

2.2 Time-Window Flexibility in Delivery Planning

The delivery literature provides a second relevant stream. In attended home delivery, retailers offer time windows to customers and manage the trade-off between customer convenience, delivery cost, and operational flexibility. Reviews of e-fulfillment and multi-channel distribution provide the broader logistics background (Agatz et al., 2008). Early work on consumer-direct grocery delivery and attended home delivery shows that time-slot availability, incentives, and delivery charges can influence the operational feasibility and cost of delivery

plans (Campbell and Savelsbergh, 2005, 2006; Agatz et al., 2011). Dynamic pricing studies then extend this perspective by linking delivery-slot decisions to customer choice and routing-cost consequences (Asdemir et al., 2009; Yang et al., 2016; Klein et al., 2019; Yang and Strauss, 2017). Strauss et al. (2021) study flexible delivery time slots, where customers accept uncertainty about the exact regular time window in exchange for a reduced delivery charge or another incentive. Their model combines customer choice behaviour, dynamic pricing, and routing-cost implications. This stream is relevant because it shows that temporal flexibility has operational value and can be treated as a decision-relevant resource.

Adjacent work in vehicle routing and delivery-time-window planning studies the spatial and temporal consequences of delivery choices in more detailed routing settings. Classical work on distribution with time windows and the vehicle routing problem with time windows provides the routing background for this boundary (Daganzo, 1987; Savelsbergh, 1985; Solomon, 1987; Bräysy and Gendreau, 2005a,b). Dynamic and stochastic vehicle-routing studies further show how richer formulations incorporate uncertainty, online decisions, and multi-period routing information (Dror et al., 1989; Bent and Van Hentenryck, 2004; Hvattum et al., 2006; Pillac et al., 2013; Ulmer et al., 2018). More recent same-day and forecast-order delivery models illustrate the additional data and routing logic required when the planner moves beyond aggregate temporal smoothing (Ulmer, 2020; Abdollahi et al., 2023).

The same delivery-flexibility literature also contains related work on customer acceptance and dynamic slot management. These studies reinforce the thesis' assumption that flexibility around delivery timing has operational value, even though the present model does not estimate customer choice or delivery-price responses (Ehmke and Campbell, 2014; Köhler et al., 2020; Koch and Klein, 2020).

At the same time, the thesis differs from attended-home-delivery pricing models in several important ways. The cited delivery-slot papers mainly study slot offering, acceptance, or pricing before the final schedule is fixed, whereas this thesis starts from an already existing schedule. In the attended-home-delivery setting, customers usually choose from a menu of time windows before the delivery plan is finalized. The firm may dynamically price or offer time slots to influence demand. In the present thesis, by contrast, the schedule already exists. The decision is not how to price incoming time slots, but which existing customer demand profiles should be repositioned after an initial aggregate load pattern is observed. Thus, the delivery-time-window literature is used to motivate the value of flexibility, not to define a behavioural choice model.

This distinction sharpens the research contribution. If the thesis were framed as dynamic pricing of delivery slots, it would compete with a mature attended-home-delivery literature. Framed as post-schedule temporal demand-profile rescheduling, it occupies a more specific niche. Flexibility remains central, but it is operationalized as feasible shifts around an existing schedule rather than as a menu of priced delivery products.

2.3 Resource Smoothing and Timed Truck Operations

The objective of the thesis is closest in spirit to resource leveling and overload minimization. Resource-leveling problems shift activities within feasible time ranges to smooth the utilization of renewable resources over a planning horizon. Rieck et al. (2012) formulate mixed-integer linear programming models for resource leveling and overload problems in project scheduling. Their work is highly relevant because it focuses explicitly on smoothing a resource-utilization profile over time. The resource in the present thesis is truck capacity, and the activities are customer demand profiles that can be moved within feasible shift windows.

The overload perspective is also useful. In the thesis, high aggregate demand in a given time slot may require owned trucks, long-term rented capacity, or spot-market capacity. A pure minimax objective reduces the highest load observed over the day, which can be interpreted as reducing the required capacity level at the most congested point. Alternatively, an overload objective could penalize demand above a threshold, such as owned-fleet capacity. The choice between minimax peak minimization and threshold-based overload minimization should depend on the managerial interpretation. If the main question is how many trucks are needed at the busiest moment, minimax is appropriate.

Minimax resource allocation provides an additional theoretical link. Klein et al. (1993) study resource allocation problems with a minimax objective, where the goal is to minimize the largest term across a set of activities under resource substitution possibilities. Although the setting differs from time-slot delivery scheduling, the paper provides a useful high-level justification for minimax thinking in resource allocation. The present thesis similarly seeks a balanced allocation of demand over time, because a plan with the same total demand can require very different peak capacity depending on how overlapping profiles are positioned.

Ready-mixed concrete delivery is the strongest operational analogue in the transportation literature. Concrete delivery involves time-sensitive, multi-trip deliveries under strict timing constraints. Yan et al. (2008) integrate production scheduling and truck dispatching of ready-mixed concrete using a mixed-integer network-flow model with side constraints. Kinable et al. (2014) define the Concrete Delivery Problem as a combined scheduling and routing problem with heterogeneous vehicles, production centers, construction sites, time windows, and strict sequencing constraints. Schmid et al. (2010) develop hybrid metaheuristics for ready-mixed concrete delivery and show that exact methods can be combined with large-neighborhood search for complex operational

instances.

These papers demonstrate that timed delivery systems can create difficult capacity and synchronization problems. Ready-mixed concrete is not identical to the B2B setting of the thesis, but it shares several features: demand is time-dependent, service windows matter, and poor temporal alignment can increase truck requirements. Additional studies on ready-mixed concrete distribution provide complementary support from modelling, decision-support, and supply-chain scheduling perspectives (Asbach et al., 2009; Matsatsinis, 2004; Naso et al., 2007). Dispatcher-oriented and industry-motivated studies provide further evidence that truck timing and dispatching are central operational concerns in this setting (Feng et al., 2004; Durbin and Hoffman, 2008; Lin et al., 2010).

The concrete-delivery papers typically solve richer problems that include routing, truck assignment, plant selection, travel times, multiple deliveries per customer, and strict unloading constraints. The present thesis deliberately abstracts from these dimensions and treats the aggregate number of trucks required per time slot as the key resource profile. This makes the model less operationally detailed than concrete-delivery dispatching models, but more aligned with the available data and with the thesis objective of temporal capacity smoothing.

2.4 Recoverable Rescheduling and Research Gap

The deterministic base model assumes that selected profile movements can be implemented. Operationally, however, some proposed movements may become unavailable after the first plan is created. This can happen because a customer cannot accommodate the new time, because internal staffing or dock constraints change, or because the planner decides not to execute a movement after further operational checks. The important point for this thesis is not to de-

velop a broad behavioural implementation model, but to ask whether the same minimax logic can be reused after some first-stage movements are no longer available.

The recoverable-robustness literature provides the conceptual foundation for this focused extension. van den Akker et al. (2018) combine two-stage planning and recoverable robustness in a scheduling setting, separating an initial plan from a limited recovery action after uncertainty is realized. This structure is useful for the thesis because it preserves the deterministic minimax model as the analytical core. The model first generates a peak-smoothing schedule, then a second optimization step can be solved with part of the first-stage plan fixed and only a limited set of additional movements allowed. Broader robust discrete-optimization work provides related background for protecting or repairing decisions under uncertainty (Kouvelis and Yu, 1997). The extension therefore remains a rescheduling problem rather than becoming a separate uncertainty-planning thesis.

The literature reviewed above supports the thesis from complementary directions. Rescheduling and schedule-stability research provide the conceptual logic: an existing plan can be revised, but the revision should be evaluated against both performance and deviation from the original schedule. Resource-leveling and minimax allocation provide the closest mathematical analogue: activities or profiles can be positioned in time to smooth a resource-utilization curve. Timed-delivery and ready-mixed-concrete research provide the operational motivation: delivery systems with time-dependent demand can face synchronization problems and costly capacity peaks. Recoverable rescheduling provides a focused implementation extension without changing the core deterministic model.

Table 1: Synthesis of literature streams and positioning of the thesis

Literature stream	Main contribution to this thesis	Remaining limitation for this problem
Rescheduling and schedule stability	Provides the logic of revising an existing plan while considering efficiency and stability.	Usually focuses on production or shop-floor environments rather than temporal truck-demand profiles.
Resource leveling and minimax allocation	Provides the closest mathematical analogy for smoothing a resource-utilization profile over time.	Usually does not model customer-specific delivery profiles or implementation constraints.
Delivery time-window flexibility	Shows that temporal flexibility has operational value.	Often studies slot offering or dynamic pricing before the final schedule is fixed.
Timed delivery and ready-mixed concrete	Demonstrates that timed truck operations can create capacity and synchronization problems.	Richer models usually include routing, dispatching, plants, travel times, and vehicle assignment.
Recoverable rescheduling	Supports a second-stage correction after part of an initial plan cannot be implemented.	Existing applications do not directly address B2B truck-demand profile smoothing.

The resulting research gap can be stated as follows. Existing studies either focus on rescheduling in production environments, resource leveling in project scheduling, flexible delivery slot pricing in attended home delivery, or integrated routing and dispatching in ready-mixed concrete delivery. They do not directly address the narrower problem of rescheduling fixed customer demand profiles across a discrete daily timeline in order to minimize peak concurrent truck requirements under limited customer flexibility. This thesis contributes to this gap

by developing and testing a minimax rescheduling model that explicitly links temporal demand smoothing with schedule-disruption control and then examines whether limited recoverable rescheduling can restore performance when part of the first-stage plan cannot be implemented.

3 Problem Description

This chapter translates the research objective into a concrete planning problem. It defines the planning context, explains the concepts of demand profiles and profile positions, gives a small illustrative example, and then states the flexibility, disruption, data, and boundary assumptions used in the proof-of-concept study.

3.1 Planning Context and Demand Profiles

The planning problem considered in this thesis concerns a single delivery day that is divided into equal-length time slots. In the generated instance set used for the numerical study, the planning horizon covers the working day from 08:00 to 18:00 and uses 15-minute intervals. Each customer request is active over a consecutive sequence of time slots and contributes an integer number of trucks to each active slot. The contribution is not necessarily constant over the customer's active period. Instead, a request can have a profile shape that increases, peaks, and then decreases.

From an operational perspective, the relevant output of the current data is the aggregate load curve over the day. For each time slot, the load is the sum of all customer contributions that are active in that slot. If many customers are active around the same time, the aggregate load can exceed the capacity level that the firm would prefer to cover with its own fleet or with predictable long-term capacity. Conversely, less popular periods may remain underutilized. The planner therefore has an incentive to move some demand profiles away from

the busiest intervals, provided that these changes remain acceptable from a customer-service perspective.

In a complete delivery-planning system, customer location, travel time, service time, vehicle assignment, and route position would also be relevant. These elements are deliberately abstracted away in the present thesis. A truck demand profile is treated as a time-dependent capacity requirement, and the model decides the final temporal placement of that requirement. This abstraction is appropriate for the current stage because the data support time-slot balancing but not spatial route construction.

Three concepts define the problem. The first concept is the demand profile. A demand profile describes the internal temporal pattern of a customer's truck requirement. For example, a profile may require relatively little capacity at the beginning and end of the active period and more capacity in the middle. This profile is fixed in shape. The model does not change the internal demand pattern of a customer.

The second concept is the profile position. The same demand profile can be placed at different start times in the daily planning horizon. If a profile starts earlier or later, its internal shape remains the same, but its contribution is shifted to different clock-time slots. Thus, the model changes the position of a customer profile, not the profile itself.

The third concept is the slot contribution. Once a profile position is chosen, the customer's contribution to each global time slot is known. The aggregate truck demand in a slot is the sum of these contributions over all customers. The optimization problem is therefore a discrete profile-placement problem: for each customer, choose one feasible final position such that the highest aggregate slot load is minimized.

3.2 Illustrative Example of Profile Placement

A small example clarifies the logic before the formal model is introduced. Consider one planning day with eight time slots and three customer profiles. Table 2 shows the profile shapes and original start slots. Each profile keeps its internal shape, but one profile may be shifted to another feasible position.

Table 2: Toy example of customer demand profiles

Customer	Profile shape	Original start slot	Interpretation
A	$[1, 2, 2, 1]$	2	Rises, peaks, and then falls.
B	$[1, 1, 2, 1]$	3	Starts flatter and peaks later.
C	$[2, 2, 1]$	4	High early contribution and then declines.

In the original schedule, the profiles overlap strongly in slots 4 and 5. The aggregate load is therefore $[0, 1, 3, 5, 5, 2, 0, 0]$, and the peak demand is 5 trucks. If customer C is shifted two slots later while A and B remain unchanged, the aggregate load becomes $[0, 1, 3, 3, 3, 3, 2, 1]$, and the peak decreases to 3 trucks. The total demand remains unchanged; only the timing of one profile changes. Table 3 summarizes this effect.

Table 3: Toy example of peak reduction through one profile shift

Schedule	Aggregate load over slots 1–8	Peak
Original schedule	$[0, 1, 3, 5, 5, 2, 0, 0]$	5
C shifted two slots later	$[0, 1, 3, 3, 3, 3, 2, 1]$	3

This example illustrates the central mechanism of the thesis. The model does not remove demand and does not split customer profiles. Instead, it chooses feasible final positions for complete profiles so that costly overlaps are reduced. The same logic is applied to larger daily instances in the numerical study.

3.3 Customer Flexibility and Disruption

Customer schedules cannot be shifted arbitrarily. Each customer has a limited set of feasible shift options. A shift of zero means that the customer remains in the original schedule. Negative shifts represent earlier placements, and positive shifts represent later placements. The feasible set may be symmetric around the original schedule, such as one or two time slots earlier or later, or it may be customer-specific if some customers are assumed to have more flexibility than others.

Even when a shift is feasible, it may still be disruptive. In practice, moving a customer may require communication, negotiation, internal re-planning, or a service concession. Moreover, a large shift is usually more disruptive than a small one. The model therefore distinguishes between three levels of disruption control. First, a binary change indicator records whether a customer is moved at all. Second, a disruption budget can limit the total number of changed customers. Third, a secondary objective can minimize the total amount of rescheduling measured by the absolute shift size.

This interpretation is consistent with the rescheduling literature, which emphasizes that a revised schedule should be evaluated not only by its efficiency but also by its stability and deviation from the original plan (Wu et al., 1993; Rangsaritratsamee et al., 2004). In this thesis, stability is represented by limited changes relative to the baseline schedule.

3.4 Data Structure, Boundaries, and Assumptions

The numerical study uses generated but empirically motivated planning instances. The generation logic creates multiple daily instances, each containing a set of customer requests. For every request, the dataset records the day, customer identifier, original start time, duration in 15-minute slots, duration in minutes, and the slot-level truck values that form the customer's active profile.

The instance set is appropriate for the purpose of this thesis because the objective is to develop and test the rescheduling logic under realistic and controlled planning conditions. The generator creates 30 daily instances, each with a working-day horizon from 08:00 to 18:00, 15-minute time intervals, and a random number of customers per day. Customer profiles are generated as rise-and-fall demand shapes, which creates time-dependent overlap patterns and makes peak smoothing meaningful. The generated nature of the data is therefore acknowledged, but it is not treated as the central limitation of the thesis; the main analytical question is whether the model can smooth realistic temporal demand profiles.

Table 4: Main data elements used by the rescheduling model

Data element	Role in the model
Day index	Identifies the daily instance to be optimized independently.
Customer identifier	Defines the set of customers included in one planning day.
Original start time	Determines the baseline position of the customer's demand profile.
Duration slots	Determines the number of consecutive time slots in the active profile.
Slot-level truck values	Defines the internal shape of the customer demand profile.
Planning horizon	Defines the set of global time slots into which profiles can be placed.
Capacity thresholds	Provide optional reference levels for owned, long-term rented, and spot-market capacity analysis.

The proof-of-concept setting is intentionally stylized. It is sufficient for analysing peak reduction, schedule disruption, and the effect of flexibility limits. It is not sufficient for analysing route feasibility, driver schedules, depot dispatching, or exact transportation costs. These aspects would require additional data. The current thesis therefore focuses on the first analytical layer: whether limited rescheduling of customer demand profiles can smooth aggregate truck demand.

The deterministic base model relies on the following assumptions:

A1. The planning problem is solved independently for each day.

A2. Time is discretized into equal-length intervals.

A3. Each customer has one fixed demand profile that remains internally unchanged.

A4. Each customer must receive exactly one final profile position.

A5. Feasible shifts are known before optimization and always include the original position.

A6. The aggregate truck requirement in a slot is the sum of all customer slot contributions.

A7. The primary objective is to minimize the maximum aggregate truck requirement over the day.

A8. Schedule disruption is controlled through a change indicator, a disruption budget, secondary objectives, or a combination of these mechanisms.

A9. Recoverable rescheduling is evaluated only after the deterministic model has established the first-stage smoothing potential.

These assumptions deliberately narrow the problem. They allow the thesis to formulate a clear mixed-integer optimization model and to test the value of temporal rescheduling before adding routing complexity or company-specific execution rules.

4 Mathematical Model

This section develops the deterministic minimax rescheduling model. The formulation follows the problem logic described above: each customer profile is assigned exactly one feasible shift, the resulting slot-level contributions are summed into an aggregate load curve, and the maximum load over the day

is minimized. The model is then extended with disruption control, optional external-capacity cost logic, and a recoverable second-stage rescheduling step.

4.1 Notation and Decision Variables

Table 5: Sets and indices

Symbol	Description
I	Set of customers scheduled on the considered day.
T	Set of discrete time slots in the daily planning horizon.
K_i	Set of feasible shift options for customer $i \in I$.
i	Customer index.
t	Time-slot index.
k	Shift-option index, measured in numbers of time slots.
u	Relative position within a customer demand profile.

Table 6: Parameters

Symbol	Description
a_i	Original start slot of customer i.
m_i	Number of active slots in the demand profile of customer i.
p_{iu}	Truck demand of customer i at relative profile position u, where $u = 1, \ldots, m_i$.
q_{itk}	Truck contribution of customer i to slot t if shift $k \in K_i$ is selected.
B	Maximum number of customers allowed to be rescheduled.
Δ_i	Maximum absolute shift allowed for customer i, measured in time slots.
c_i	Fixed disruption penalty if customer i is moved.
h_i	Variable disruption penalty per shifted time slot for customer i.
C^{own}	Own-fleet capacity threshold.
C^{long}	Long-term rented capacity available in addition to the own fleet.
c^{long}	Cost coefficient for using long-term rented capacity in a time slot.
c^{spot}	Cost coefficient for using spot-market capacity in a time slot.
B^R	Maximum number of additional customers allowed to be changed in the recovery stage.
F	Set of customers whose recovery-stage positions are fixed.
R	Set of customers eligible for second-stage recovery movements.
$\bar{k}_i$	Fixed recovery-stage shift for customer $i \in F$.

The feasible shift set can be constructed from the daily planning horizon and the maximum allowed shift. For example, if customer i may be moved at most Δ_i slots earlier or later, then

$$K_i = \{k \in \mathbb{Z} : |k| \leq \Delta_i,\ 1 \leq a_i + k,\ a_i + k + m_i - 1 \leq |T|\}. \tag{1}$$

This definition ensures that the shifted profile remains inside the planning horizon. The option $k = 0$ corresponds to the original schedule and is included whenever the original schedule is feasible.

The parameter q_{itk} can be precomputed before optimization. It converts the relative customer profile into an absolute time-slot contribution under shift k:

$$q_{itk} = \begin{cases} p_{iu}, & \text{if } t = a_i + k + u - 1 \text{ for some } u \in \{1, \ldots, m_i\}, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

This distinction is important because p_{iu} describes the internal profile shape, whereas q_{itk} describes the realized contribution of that profile to the global daily timeline.

The decision variables are grouped by the subproblem in which they are used. This separation is important because the deterministic base model, the disruption-control extension, the optional capacity-tier formulation, and the recovery model do not all use the same variables.

Table 7: Decision variables by model component

Symbol	Description
Deterministic minimax base model	
x_{ik}	Binary variable equal to 1 if customer i is assigned shift k, and 0 otherwise.
L_t	Aggregate truck demand in time slot t.
Z	Maximum aggregate truck demand over all time slots.
Disruption-control variables	
y_i	Binary variable equal to 1 if customer i is moved away from the original schedule, and 0 otherwise.
s_i	Absolute shift size assigned to customer i.
Optional capacity-tier extension	
G_t	Long-term rented capacity used above own-fleet capacity in time slot t.
H_t	Spot-market capacity used above own-fleet and long-term rented capacity in time slot t.
Recoverable rescheduling extension	
x_{ik}^R	Binary recovery-stage variable equal to 1 if eligible customer i is assigned shift k in the recovery solution.
y_i^R	Binary recovery-stage variable equal to 1 if eligible customer i is changed during recovery.
Z^R	Maximum aggregate truck demand in the recovery solution.

4.2 Deterministic Minimax Base Model

The pure base model minimizes the highest aggregate truck demand over the day. It does not yet restrict the number of changed customers; disruption control is introduced in the following subsection. The base model is

$$\min \quad Z \tag{3}$$

$$\text{s.t.} \quad \sum_{k \in K_i} x_{ik} = 1 \qquad \forall i \in I, \tag{4}$$

$$L_t = \sum_{i \in I} \sum_{k \in K_i} q_{itk} x_{ik} \qquad \forall t \in T, \tag{5}$$

$$L_t \leq Z \qquad \forall t \in T, \tag{6}$$

$$x_{ik} \in \{0, 1\} \qquad \forall i \in I,\ k \in K_i, \tag{7}$$

$$L_t \geq 0 \qquad \forall t \in T, \tag{8}$$

$$Z \geq 0. \tag{9}$$

Constraint (4) is the shift-assignment constraint. It ensures that every customer receives exactly one final schedule option. Importantly, it does not require every customer to be moved. If $0 \in K_i$, then $x_{i0} = 1$ means that customer i remains unchanged. The constraint prevents two invalid outcomes: a customer disappearing from the plan, and a customer being assigned to multiple final positions.

Constraint (5) constructs the aggregate load curve. For a given slot t, only the selected shift option of each customer contributes to the sum because exactly one x_{ik} equals 1. Constraint (6) links each slot load to the peak variable Z. Because the objective minimizes Z, the optimal value of Z equals the highest point of the aggregate load curve.

The base model is a mixed-integer linear program. It is linear because the contribution parameters q_{itk} are precomputed and the decision variables only select among feasible profile placements. This is computationally attractive

for the proof-of-concept stage and keeps the model transparent for managerial interpretation.

4.3 Disruption Control and Lexicographic Refinement

The pure minimax model may reduce the peak by moving many customers. To represent operational disruption, the model can be extended with a change indicator and a disruption budget. The change indicator should be linked exactly to the shift decision:

$$y_i = \sum_{k \in K_i \setminus \{0\}} x_{ik} \qquad \forall i \in I, \tag{10}$$

$$\sum_{i \in I} y_i \leq B, \tag{11}$$

$$y_i \in \{0, 1\} \qquad \forall i \in I. \tag{12}$$

Equation (10) is stronger and cleaner than a one-sided activation constraint. It ensures that $y_i = 1$ if customer i receives any nonzero shift and $y_i = 0$ if the customer remains in the original position. If $0 \in K_i$ for every customer, the same logic can equivalently be written as $y_i = 1 - x_{i0}$. Constraint (11) limits the number of customers whose schedules may be changed.

The absolute shift size of each customer can be represented as

$$s_i = \sum_{k \in K_i} |k| x_{ik} \qquad \forall i \in I. \tag{13}$$

The variable s_i is zero for unchanged customers and positive for moved customers. It can be used either as a secondary objective or as part of a weighted disruption penalty.

A pure minimax objective may produce several alternative solutions with the same peak value. Some of these solutions may move more customers than

necessary. To avoid this, the thesis uses a lexicographic interpretation as the preferred disruption-control approach. The optimization is solved in three steps.

First, solve the deterministic minimax model and denote the optimal peak by Z^*:

$$Z^* = \min Z. \tag{14}$$

Depending on the experimental setting, this first step can be solved either without a disruption budget or with a fixed budget B if the planner wants to impose an operational limit from the start.

Second, restrict the peak to remain optimal and minimize the number of changed customers:

$$\min \quad \sum_{i \in I} y_i \tag{15}$$

$$\text{s.t.} \quad Z \leq Z^*, \tag{16}$$

with all constraints from the base model and the exact change-indicator constraints. Let the optimal number of changed customers be Y^*.

Third, restrict both the peak and the number of changed customers and minimize the total shift magnitude:

$$\min \quad \sum_{i \in I} s_i \tag{17}$$

$$\text{s.t.} \quad Z \leq Z^*, \tag{18}$$

$$\sum_{i \in I} y_i \leq Y^*, \tag{19}$$

again with all constraints from the base model and the disruption definitions. This hierarchy reflects the managerial logic of the problem: peak reduction is the primary operational goal, but among equally good peak-reducing plans, the least disruptive plan is preferred.

A weighted objective can be used as an alternative:

$$\min \ \alpha Z + \beta \sum_{i \in I} c_i y_i + \gamma \sum_{i \in I} h_i s_i, \tag{20}$$

where α, β, and γ are non-negative weights. This form is useful for sensitivity analysis or when managerial cost parameters are available. At the current stage, however, the lexicographic formulation is preferable because it avoids arbitrary calibration of penalty weights before empirical cost data are available.

4.4 Capacity-Tier and Recoverable Extensions

The minimax objective reduces the required capacity at the busiest moment. In some applications, however, the planner may care more directly about the cost of exceeding certain capacity thresholds. For example, demand up to C^{own} may be covered by owned trucks, demand above that level may require long-term rented capacity, and demand above $C^{\text{own}} + C^{\text{long}}$ may require spot-market capacity.

A transparent linear tier formulation introduces nonnegative variables G_t and H_t:

$$L_t \leq C^{\text{own}} + G_t + H_t \qquad \forall t \in T, \tag{21}$$

$$0 \leq G_t \leq C^{\text{long}} \qquad \forall t \in T, \tag{22}$$

$$H_t \geq 0 \qquad \forall t \in T. \tag{23}$$

Here, G_t represents capacity used in the long-term rented tier, while H_t represents capacity above the own-fleet and long-term rented thresholds. The planner could then minimize a cost expression such as

$$\min \ c^{\text{long}} \sum_{t \in T} G_t + c^{\text{spot}} \sum_{t \in T} H_t + \beta \sum_{i \in I} y_i + \gamma \sum_{i \in I} s_i. \tag{24}$$

This formulation avoids ambiguity about double-counting overload above multiple thresholds. It should nevertheless be interpreted cautiously. It requires assumptions about the cost of capacity tiers and about whether the planner pays per time slot, per truck, or per daily capacity commitment. Therefore, it is best treated as a sensitivity or managerial extension rather than as the central base model.

The deterministic model produces a first-stage rescheduling plan. In practice, not every planned movement may remain executable after additional operational information becomes available. A customer may not be able to accommodate the new time, a dock or staffing constraint may change, or the planner may decide to preserve part of the original schedule after a feasibility check. The recoverable extension asks whether a limited second-stage optimization can restore part of the lost smoothing effect without changing the structure of the core model.

Let $\hat{k}_i$ denote the first-stage shift selected for customer i. After implementation information is observed, some first-stage shifts are fixed and others are unavailable. Let F denote the set of customers whose final position is fixed in the recovery stage, and let $R \subseteq I \setminus F$ denote the customers that remain eligible for recovery movements. For each fixed customer $i \in F$, let $\bar{k}_i$ be the shift that must be used in the recovery stage. The fixed load contribution in slot t is then

$$\bar{L}_t = \sum_{i \in F} q_{it\bar{k}_i} \qquad \forall t \in T. \tag{25}$$

For eligible customers $i \in R$, the recovery model again chooses one feasible shift. The recovery decision variable x^R_{ik} equals one if customer i receives shift

k in the recovery solution. The recovery formulation is

$$\min \quad Z^R \tag{26}$$

$$\text{s.t.} \quad \sum_{k \in K_i} x_{ik}^R = 1 \qquad \forall i \in R, \tag{27}$$

$$\bar{L}_t + \sum_{i \in R} \sum_{k \in K_i} q_{itk} x_{ik}^R \leq Z^R \qquad \forall t \in T, \tag{28}$$

$$\sum_{i \in R} y_i^R \leq B^R, \tag{29}$$

$$x_{ik}^R \leq y_i^R \qquad \forall i \in R,\ k \in K_i \setminus \{0\}, \tag{30}$$

$$x_{ik}^R \in \{0,1\} \qquad \forall i \in R,\ k \in K_i, \tag{31}$$

$$y_i^R \in \{0,1\} \qquad \forall i \in R, \qquad Z^R \geq 0. \tag{32}$$

Here, B^R is the recovery budget, measured as the maximum number of additional customers that may be changed in the recovery step. The formulation deliberately mirrors the deterministic minimax model. The only difference is that some load is already fixed through $\bar{L}_t$, and only a restricted subset of customers is adjustable.

This extension is not a separate uncertainty model. It is a reuse of the deterministic minimax formulation after partial implementation information is known. Its purpose is to test whether the same decision-support model can support both the initial rescheduling proposal and a limited corrective action.

4.5 Summary of Modelling Choices

The mathematical model deliberately focuses on the temporal capacity-smoothing problem. Its main modelling choices are summarized in Table 8.

Table 8: Summary of modelling choices

Model element	Modelling choice
Planning unit	One delivery day solved independently.
Time representation	Discrete 15-minute time slots.
Customer representation	Fixed demand profile with a selectable final position.
Primary decision	Choose one feasible shift for each customer.
Primary objective	Minimize the maximum aggregate truck demand over the day.
Disruption control	Limit or penalize changed customers and shift magnitudes, with exact linkage between shift decisions and change indicators.
Capacity-cost extension	Optional tier formulation for owned, long-term rented, and spot-market capacity; used for managerial interpretation rather than solved as a main numerical experiment.
Recoverable extension	Re-solve the same minimax profile-placement problem for a limited set of adjustable customers after part of the first-stage plan is fixed or unavailable.
Routing scope	Excluded from the base model due to missing spatial data.

The formulation provides the basis for the numerical study. The deterministic base model, the disruption-budget variant, and the lexicographic objective logic

are used directly in the flexibility and disruption-budget experiments. The recoverable formulation is used in the second-stage recovery experiment. The capacity-tier formulation is intentionally retained as an optional managerial extension only: it explains how peak smoothing could later be translated into capacity-cost terms, but it is not solved as a separate numerical experiment because calibrated tier-cost parameters are not available in the generated instance set. The model components used in the numerical study are therefore aligned with the available data and with the thesis' main research questions.

5 Numerical Study

This chapter evaluates the minimax rescheduling model developed in Chapter 4. The purpose is to test whether limited changes in customer profile positions can reduce peak simultaneous truck demand, how this reduction depends on the amount of permitted temporal flexibility, and how the benefit changes when the number of customer movements is restricted. The chapter then studies a recoverable rescheduling extension in which the same minimax logic is applied again after part of the first-stage plan cannot be implemented.

The numerical study is organized as follows. Section 5.1 describes the data, experimental setup, and computational implementation. Section 5.2 analyzes the original load structure and the value of increasing the allowed time shift. Section 5.3 introduces disruption budgets, evaluates the efficiency-disruption frontier, and examines day-level variation. Section 5.4 analyzes whether part of the lost smoothing benefit can be restored through limited second-stage rescheduling. Section 5.5 summarizes the main numerical findings.

5.1 Data, Experimental Setup, and Implementation

The numerical study uses the customer-request dataset described in Section 3.4. The dataset contains 30 independent daily planning instances and 518 customer requests in total. Each day is solved separately, consistent with the daily planning logic introduced in Chapter 3. The planning horizon runs from 08:00 to 18:00 and is represented by 41 slot-start positions. The slot sequence therefore includes 08:00, 08:15, ..., 17:45, and 18:00. Each slot has a length of 15 minutes.

Each customer request contains an original start time, a profile duration, and a sequence of slot-level truck contributions. The implementation first reconstructs the original aggregate load curve for each day by placing all customer profiles at their original start times. It then constructs the feasible shift set K_i for each customer. The option $k = 0$ is always included and represents the original schedule. A nonzero shift is feasible if the shifted profile remains inside the planning horizon.

Table 9 summarizes the instance characteristics. The daily number of customers ranges from 12 to 24, with an average of 17.27 customers per day. The average baseline peak is 32.03 trucks, while the average slot load is 14.69 trucks. This difference already indicates that peak demand is driven by temporal concentration rather than by the average demand level alone.

Table 9: Summary of the daily planning instances

Measure	Mean	Minimum	Maximum	Total
Number of daily instances	–	–	–	30
Customer requests per day	17.27	12	24	518
Baseline peak truck demand	32.03	18	45	–
Total truck-slot demand per day	602.27	401	909	–
Mean slot load per day	14.69	9.78	22.17	–
Planning horizon slots	41	41	41	–
Slot length in minutes	15	15	15	–

Three groups of experiments are conducted. The first group varies the maximum allowed customer shift and measures the value of temporal flexibility. The second group fixes the flexibility at ± 60 minutes and varies the share of customers that may be moved, thereby measuring the trade-off between peak reduction and schedule disruption. The third group studies whether the planner can react to partial implementation of the first-stage plan through a limited second-stage recovery action. Table 10 summarizes the experimental design.

Table 10: Experimental design of the numerical study

Experiment	Varied parameter	Purpose
Flexibility sensitivity	Maximum shift of 0, 15, 30, and 60 minutes in either direction	Measures how additional time-window flexibility affects peak reduction when the model may choose any beneficial customer movement.
Disruption-budget sensitivity	Maximum changed-customer share of 0%, 10%, 20%, 30%, 50%, and 100%	Measures how much peak reduction remains when only a limited number of customer schedules may be changed.
Recoverable rescheduling extension	First-stage implementation probabilities of 80%, 60%, and 40%, and recovery budgets of 0%, 10%, and 20%	Tests whether a limited second-stage rescheduling action can restore part of the peak-reduction benefit when some first-stage movements are unavailable.

The main deterministic performance indicators are the optimized peak, the absolute peak reduction, the relative peak reduction, the number of changed customers, and the total shift magnitude. For day d, let Z_d^{base} denote the original peak and let Z_d^{opt} denote the optimized peak under a given experimental setting. The absolute peak reduction is

$$\Delta Z_d = Z_d^{\text{base}} - Z_d^{\text{opt}}, \tag{33}$$

and the relative peak reduction is

$$r_d = \frac{Z_d^{\text{base}} - Z_d^{\text{opt}}}{Z_d^{\text{base}}}. \tag{34}$$

For the disruption-budget analysis, an additional metric is used to compare limited-budget solutions with the unrestricted benchmark. Let $\overline{\Delta Z}_\beta$ denote the average absolute peak reduction under movement budget β, and let $\overline{\Delta Z}_{100\%}$ denote the average absolute peak reduction when all customers may be moved. The captured unrestricted benefit is defined as

$$\eta_\beta = \frac{\overline{\Delta Z}_\beta}{\overline{\Delta Z}_{100\%}}. \tag{35}$$

A value of $\eta_\beta = 0.50$ means that the limited-budget solution obtains half of the peak reduction achieved by the unrestricted solution. This metric is useful because it directly expresses how much of the maximum deterministic smoothing potential can be captured with a smaller rescheduling intervention.

The main deterministic setting uses a maximum shift of ± 60 minutes, corresponding to four 15-minute time slots. This setting is used as the central flexibility case because it represents a moderate form of operational rescheduling. It gives the optimizer enough room to separate overlapping demand profiles, while keeping the final delivery position close to the original schedule. In managerial terms, a shift of up to one hour can still be interpreted as an adjustment within the same broad delivery period rather than as a complete reallocation of the request. To avoid making the analysis depend on this single assumption, the flexibility sensitivity also includes 0, 15, and 30 minutes. The 60-minute case is therefore the upper end of a controlled flexibility experiment, not an isolated parameter choice.

The model was implemented in Python as a mixed-integer linear program. For each day, the implementation builds all feasible customer-shift combinations

and precomputes the slot contribution parameter q_{itk}. The binary variable x_{ik} equals one if customer i is assigned shift k, and zero otherwise. The aggregate load in time slot t is then obtained by summing the selected slot contributions over all customers. The peak variable Z is constrained to be at least as large as every slot load.

The computational implementation follows the lexicographic logic from Section 4.3. To avoid repeatedly solving three separate optimization models for each day and each experimental setting, the implementation uses a single weighted lexicographic objective:

$$\min \; W_Z Z + W_Y \sum_{i \in I} y_i + W_S \sum_{i \in I} s_i. \tag{36}$$

The three objective components correspond to the three priorities of the deterministic model. The first priority is to minimize the peak load. The second priority is to avoid unnecessary customer changes. The third priority is to minimize total shift magnitude among otherwise equivalent solutions.

The weights are chosen dynamically for each daily instance. Let Δ denote the maximum allowed absolute shift in slots for the respective experiment. The implementation sets

$$W_S = 1, \quad W_Y = |I|\Delta + 1, \quad W_Z = |I|W_Y + |I|\Delta + 1. \tag{37}$$

This weighting scheme preserves the intended priority order. A one-unit reduction in Z dominates any possible change in the number of moved customers and total shift magnitude. Similarly, a one-unit reduction in the number of moved customers dominates any possible change in total shift magnitude. The model therefore behaves like the lexicographic formulation, while remaining computa-

tionally efficient for the experiment grid.

For a requested movement budget β, the daily customer budget is computed as

$$B_d = \lfloor \beta |I_d| \rfloor, \tag{38}$$

where I_d is the set of customers on day d. The actual number of moved customers may be smaller than B_d because the objective does not use remaining budget unless this improves the lexicographic solution.

5.2 Baseline Load Structure and Customer Flexibility

The original schedules show a clear peak-load pattern. The average baseline peak equals 32.03 trucks, whereas the average load over all slots equals 14.69 trucks. Thus, the average peak is more than twice as high as the average slot load. This confirms that the main issue is not the daily amount of truck demand alone, but the temporal overlap of customer profiles.

Figure 1 shows the distribution of baseline peak times across the 30 daily instances. The most frequent peak time is 14:00. Other frequent peak times are located around 12:15, 12:30, and 13:30. The concentration of peaks around the middle and early afternoon part of the day provides a clear motivation for the rescheduling model: if selected profiles can be shifted away from these congested intervals, the maximum simultaneous capacity requirement can be reduced.

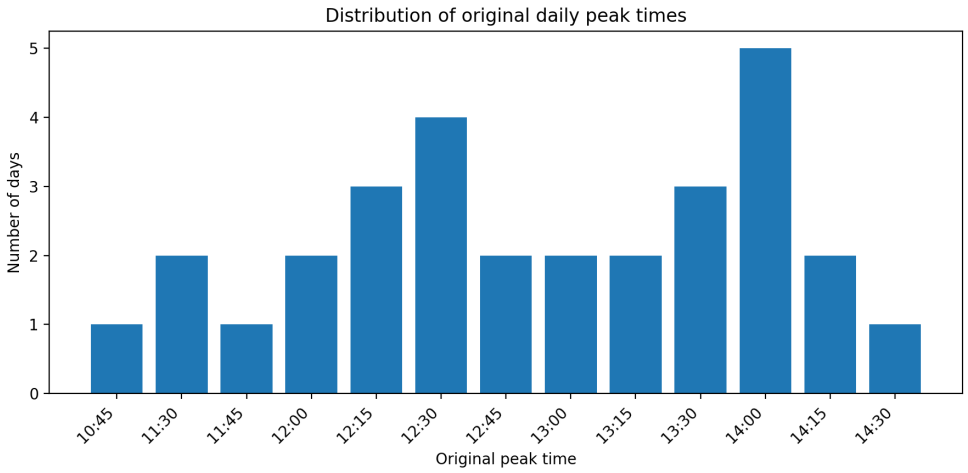


Figure 1: Distribution of original daily peak times across the planning instances

Figure 2 compares the average original load curve with the average optimized load curve under unrestricted ± 60 minute flexibility. The optimized curve is visibly flatter. The model reduces the highest part of the average load profile and redistributes demand toward less congested parts of the horizon. The total truck-slot demand remains unchanged because every customer profile is still assigned exactly one final position.

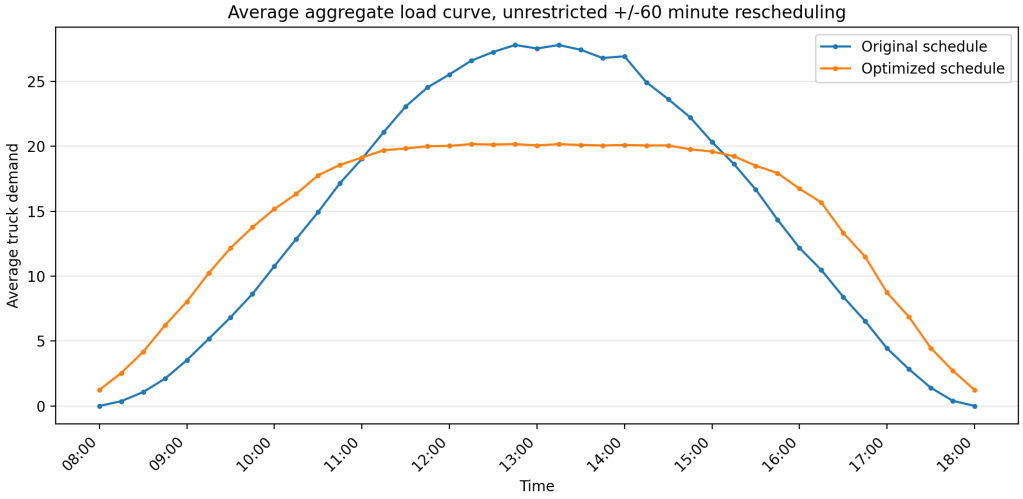


Figure 2: Average aggregate load curve before and after unrestricted deterministic rescheduling with ± 60 minutes flexibility

The unrestricted ± 60 minute model reduces the average daily peak from 32.03 trucks to 20.50 trucks. This corresponds to an average absolute reduction of

11.53 trucks and an average relative reduction of 36.07%. This result establishes the deterministic smoothing potential of the model. At the same time, the unrestricted solution changes 12.83 customers per day on average, corresponding to 74.27% of customers. This motivates the disruption-budget analysis below.

The first sensitivity analysis studies how the maximum allowed shift affects the achievable peak reduction. Four flexibility levels are considered: no movement, ± 15 minutes, ± 30 minutes, and ± 60 minutes. No explicit movement budget is imposed in this experiment. The model may therefore use as many movements as needed, while still avoiding unnecessary changes through the lexicographic objective.

Table 11 reports the average results across the 30 daily instances. Without flexibility, the optimized peak equals the original peak. With only ± 15 minutes of flexibility, the average peak reduction already reaches 11.88%. With ± 30 minutes, the reduction increases to 22.85%. With ± 60 minutes, the reduction reaches 36.07%.

Table 11: Average results under different flexibility levels without an explicit movement budget

Maximum shift (minutes)	Optimized peak (trucks)	Absolute reduction (trucks)	Relative reduction (%)	Changed customers per day	Changed share (%)
0	32.03	0.00	0.00	0.00	0.00
15	28.27	3.77	11.88	7.13	41.73
30	24.77	7.27	22.85	10.50	61.37
60	20.50	11.53	36.07	12.83	74.27

Figure 3 visualizes the same relationship. The effect of flexibility is monotonic: as the feasible shift range expands, the model has more options to separate

overlapping profiles. The first 15 minutes of flexibility already produce a visible improvement. Additional flexibility from 15 to 30 minutes and from 30 to 60 minutes remains valuable, although the larger flexibility levels also involve a higher share of changed customers.

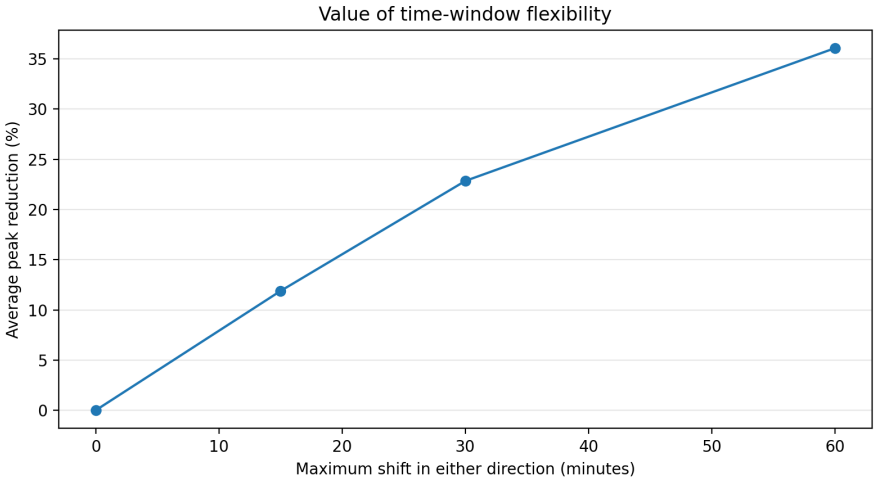


Figure 3: Average peak reduction by maximum allowed customer shift

This experiment shows that customer flexibility functions as an operational resource. Even a small amount of temporal flexibility creates peak-reduction potential. However, if movement is unrestricted, the model uses a large number of customer changes to obtain the best possible peak. The next experiment therefore asks how much of this benefit remains when the planner limits the number of customers that may be moved.

5.3 Disruption-Budget and Day-Level Analysis

The disruption-budget experiment fixes the maximum shift at ± 60 minutes and varies the maximum share of customers that may be changed. The tested movement budgets are 0%, 10%, 20%, 30%, 50%, and 100%. The 100% case represents the unrestricted benchmark under ± 60 minutes flexibility.

Table 12 reports the main results. The average optimized peak decreases as the movement budget increases. A 10% requested budget reduces the average

peak from 32.03 to 28.93 trucks. A 20% budget reduces it to 26.27 trucks. A 50% budget reduces it to 22.47 trucks. The unrestricted 100% case reaches 20.50 trucks.

Table 12: Peak reduction and schedule disruption under movement budgets

Requested budget (%)	Actual moved customers (%)	Optimized peak (trucks)	Absolute reduction (trucks)	Relative reduction (%)	Captured benefit (%)	Total shift per day (min)
0	0.00	32.03	0.00	0.00	0.00	0.0
10	7.42	28.93	3.10	9.47	26.88	59.0
20	17.49	26.27	5.77	17.96	50.00	141.5
30	26.29	24.63	7.40	23.11	64.16	223.0
50	44.14	22.47	9.57	29.87	82.95	374.5
100	74.27	20.50	11.53	36.07	100.00	625.5

The actual moved-customer share is below the requested budget in several cases. This is expected. First, the customer budget is rounded down to an integer number of customers for each day. Second, the lexicographic objective does not move customers unless the movement improves the solution according to the objective hierarchy. The unused budget is therefore not a modelling error; it is a useful property of the formulation.

Figure 4 shows the average peak reduction by budget level. The relationship is increasing but not linear. The first movements create a large part of the total benefit, while later movements deliver smaller additional improvements. This pattern is central for the interpretation of the model because it shows that the value of rescheduling is concentrated in a subset of particularly relevant customer profiles.

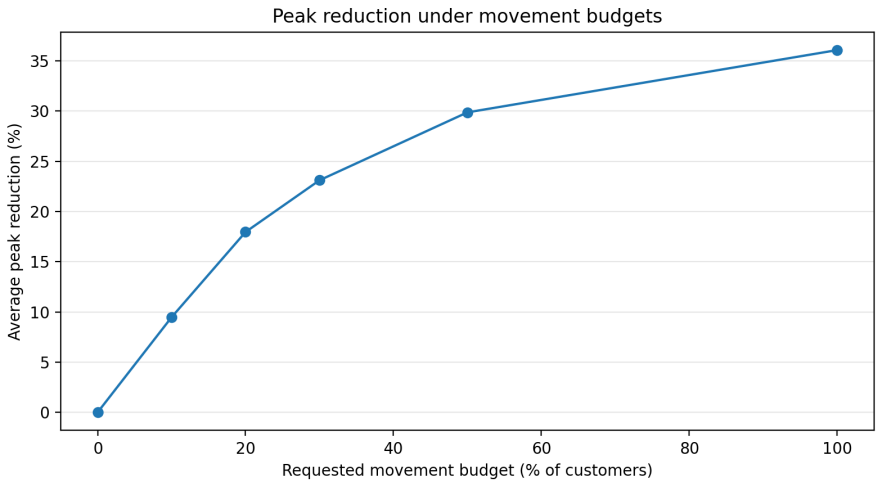


Figure 4: Average peak reduction by allowed changed-customer budget under ± 60 minutes flexibility

The same pattern can be seen in the efficiency-disruption frontier in Figure 5. The horizontal axis shows the actual share of customers moved, while the vertical axis shows the average peak reduction. A 20% requested movement budget changes 17.49% of customers on average and achieves a 17.96% peak reduction. A 50% requested budget changes 44.14% of customers and achieves a 29.87% reduction. The unrestricted solution achieves the highest reduction, but it requires a much broader intervention in the schedule.

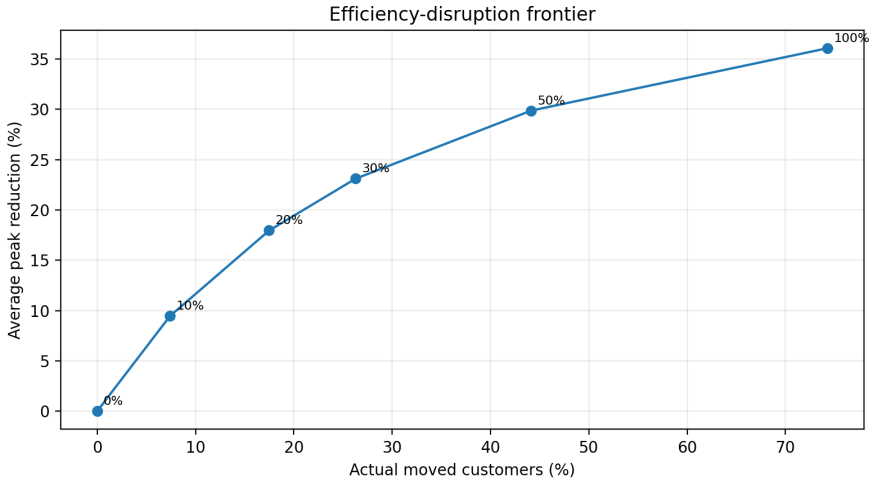


Figure 5: Efficiency-disruption frontier under ± 60 minutes flexibility

Figure 6 translates the budget results into captured unrestricted benefit. A

10% requested movement budget captures 26.88% of the unrestricted peak-reduction benefit. A 20% budget captures 50.00%. A 50% budget captures 82.95%. This finding is one of the central numerical results of the thesis: a limited number of carefully selected customer movements can obtain a large share of the total deterministic smoothing potential.

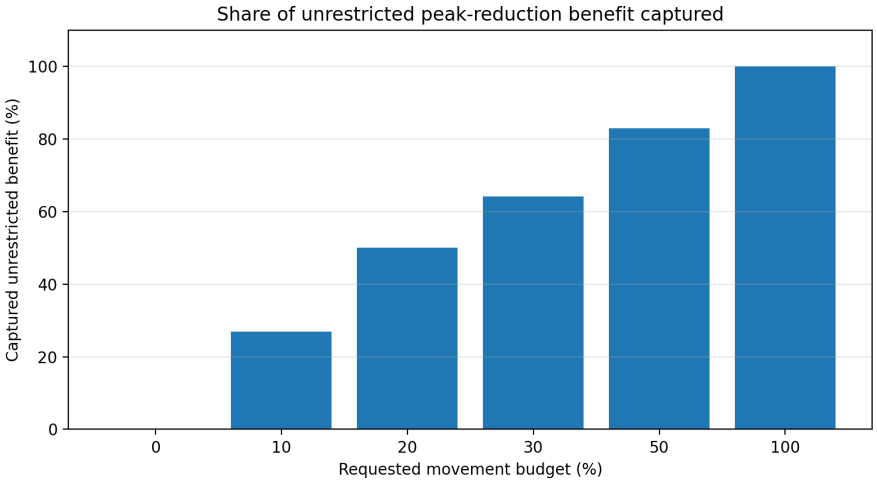


Figure 6: Share of unrestricted peak-reduction benefit captured by limited movement budgets

Table 13 makes the diminishing-return pattern explicit. The move from 0% to 10% budget adds 9.47 percentage points of average peak reduction while changing 7.42 additional percentage points of customers. This corresponds to 1.28 peak-reduction percentage points per additional changed-customer percentage point. In contrast, the move from 50% to 100% budget adds only 6.20 percentage points of peak reduction while changing 30.13 additional percentage points of customers, corresponding to 0.21 peak-reduction percentage points per additional changed-customer percentage point.

Table 13: Marginal effect of increasing the movement budget

Budget increase	Additional moved share (pp)	Additional peak reduction (pp)	Additional captured benefit (pp)	Reduction per moved share ratio
0% to 10%	7.42	9.47	26.88	1.28
10% to 20%	10.07	8.48	23.12	0.84
20% to 30%	8.80	5.15	14.16	0.59
30% to 50%	17.85	6.76	18.79	0.38
50% to 100%	30.13	6.20	17.05	0.21

The marginal analysis gives the budget results a clear managerial interpretation. The first set of movements is highly efficient because it targets profiles that contribute strongly to the peak. Further movements still improve the peak, but the additional reduction becomes less attractive relative to the additional disruption. This supports the use of a movement budget as a practical control mechanism rather than solving only the unrestricted minimax model.

The average results show the overall trend, but it is also important to examine whether the model works consistently across daily instances. Table 14 reports the distribution of daily relative peak reductions for selected movement budgets. All 30 days improve under every positive budget level. Under a 10% budget, daily reductions range from 3.33% to 15.79%. Under a 20% budget, they range from 6.67% to 27.59%. Under the unrestricted 100% case, they range from 27.27% to 41.38%.

Table 14: Distribution of daily peak reductions under selected movement budgets

Budget (%)	Mean (%)	Minimum (%)	25th pct. (%)	Median (%)	75th pct. (%)	Maximum (%)
10	9.47	3.33	6.96	9.10	11.78	15.79
20	17.96	6.67	16.05	17.44	20.34	27.59
30	23.11	13.33	20.56	23.21	25.93	31.03
50	29.87	20.00	27.03	30.99	33.09	38.24
100	36.07	27.27	32.66	36.85	39.43	41.38

Figure 7 visualizes the same day-level variation. The boxplots shift upward as the movement budget increases, which confirms that the improvement is not driven by only a few exceptional days. At the same time, the spread of the boxplots indicates that the magnitude of improvement depends on the original overlap structure of each day. Some days contain peaks that can be reduced strongly with a small number of moves, while other days require broader intervention to achieve comparable smoothing.

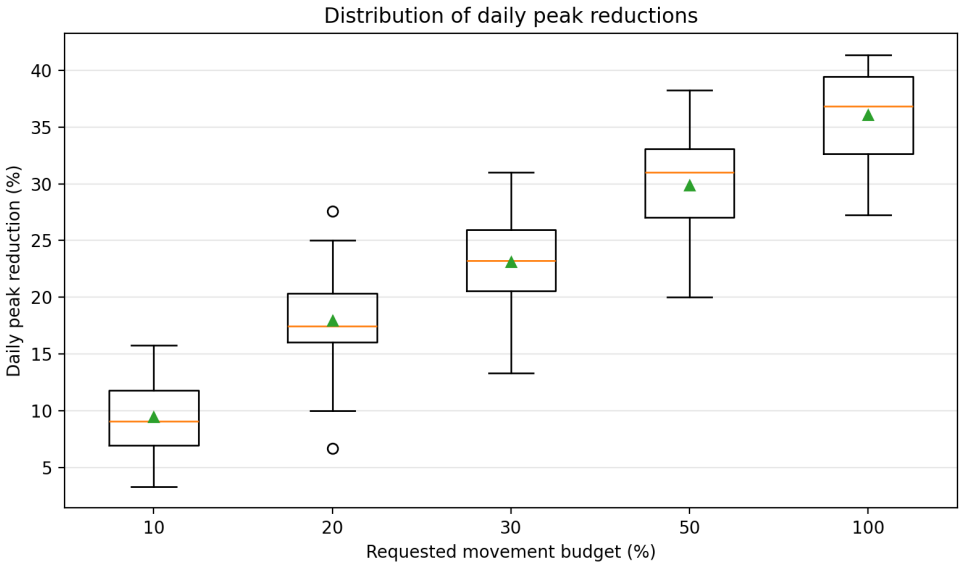


Figure 7: Distribution of daily peak reductions by movement budget

This day-level analysis strengthens the average findings. The model does not only perform well on average; it generates positive peak reductions for every daily instance in the experiment whenever movement is allowed. The exact benefit differs across days, which is useful information for implementation: the model can also be used diagnostically to identify days on which rescheduling is especially valuable.

5.4 Recoverable Rescheduling Extension

The deterministic experiments show what can be achieved when the first-stage rescheduling plan is fully implemented. The recoverable extension asks a narrower operational question: if part of this first-stage plan becomes unavailable, can a limited second-stage rescheduling action restore a meaningful share of the lost peak-reduction benefit? This section keeps the deterministic minimax model as the central model and uses recovery only as an implementation-oriented extension.

The recovery experiment follows a two-step logic. First, the deterministic disruption-budget model creates a first-stage plan. Second, only a subset of the nonzero first-stage movements is assumed to remain implementable. Implemented movements are fixed at their planned positions, non-implemented movements return to the original position, and the recovery model may move a limited number of customers that were not part of the first-stage movement set. The recovery budget therefore represents additional planning effort after the first-stage outcome is known.

This logic is intentionally recoverable rather than fully conservative. A fully conservative approach would try to protect the first-stage plan against many possible implementation outcomes before they occur. The recoverable approach instead accepts that some information becomes available only after the initial plan is proposed and then solves a limited corrective problem. This is opera-

tionally natural for delivery scheduling, where planners often revise a plan after receiving updated feasibility information.

The first-stage plan uses the 50% movement-budget solution from the deterministic analysis. This budget is chosen because it captures a large share of the unrestricted deterministic benefit while avoiding the broad disruption of the unrestricted solution. Three implementation probabilities are tested: 80%, 60%, and 40%. These probabilities should be interpreted as stress-test parameters for the recoverable mechanism, not as estimated behavioural response probabilities. For each day and implementation probability, 100 Monte Carlo scenarios are generated, yielding 3,000 scenario-day observations per implementation-probability setting across the 30 days. For every scenario, recovery budgets of 0%, 10%, and 20% of daily customers are compared.

The main performance measures are the realized peak reduction before recovery, the realized peak reduction after recovery, the incremental recovery value, and the recovered lost benefit. The incremental recovery value is the difference between peak reduction after and before recovery. The recovered lost benefit measures how much of the peak-reduction loss relative to full first-stage implementation is restored by the recovery step. Its reported average is calculated over scenario-day observations with a positive loss of deterministic benefit; the ratio is undefined and therefore omitted when no benefit was lost.

Table 15 reports the aggregate results. Without recovery, the average realized peak reduction is 21.84% when the implementation probability is 80%, 15.91% when it is 60%, and 10.22% when it is 40%. These values are below the deterministic 50%-budget result because not every planned movement is implemented.

Table 15: Recoverable rescheduling results under a 50% first-stage movement budget

Implementation probability	Recovery budget share	Before recovery reduction (%)	After recovery reduction (%)	Incremental value (pp)	Recovered lost benefit (%)	Recovery changes (customers/day)
40%	0%	10.22	10.22	0.00	0.00	0.00
40%	10%	10.22	15.35	5.13	27.19	1.24
40%	20%	10.22	19.51	9.29	49.69	2.80
60%	0%	15.91	15.91	0.00	0.00	0.00
60%	10%	15.91	20.33	4.42	33.44	1.17
60%	20%	15.91	23.88	7.97	61.34	2.69
80%	0%	21.84	21.84	0.00	0.00	0.00
80%	10%	21.84	25.01	3.17	41.76	0.98
80%	20%	21.84	27.79	5.96	77.22	2.48

The results show that recovery can restore a substantial part of the value lost through partial implementation. At 60% implementation probability, the pre-recovery peak reduction is 15.91%. Allowing an additional recovery budget of 10% increases the realized reduction to 20.33%, while a 20% recovery budget increases it to 23.88%. The larger recovery budget therefore adds 7.97 percentage points of peak reduction and restores approximately 61.34% of the lost deterministic benefit. This is a sizeable improvement considering that the recovery model moves only 2.69 additional customers per day on average.

The value of recovery is also visible at lower implementation levels. At 40% implementation probability, recovery with a 20% budget increases the peak reduction from 10.22% to 19.51%. This nearly doubles the realized smoothing effect. At 80% implementation probability, the pre-recovery outcome is already relatively strong, but a 20% recovery budget still increases the reduction from 21.84% to 27.79%.

Figure 8 visualizes the realized peak reduction across implementation probabilities and recovery budgets. For each combination, the 100 simulated outcomes are first averaged within each of the 30 daily instances. The shaded 95% normal-approximation confidence bands are then calculated as the overall mean of these daily averages plus or minus 1.96 times their standard er-

ror, $s_{\text{day}}/\sqrt{30}$. This day-clustered calculation respects the nesting of simulation outcomes within daily instances and reflects cross-day variation in mean performance. The curves show that second-stage recovery improves realized performance for all tested implementation probabilities.

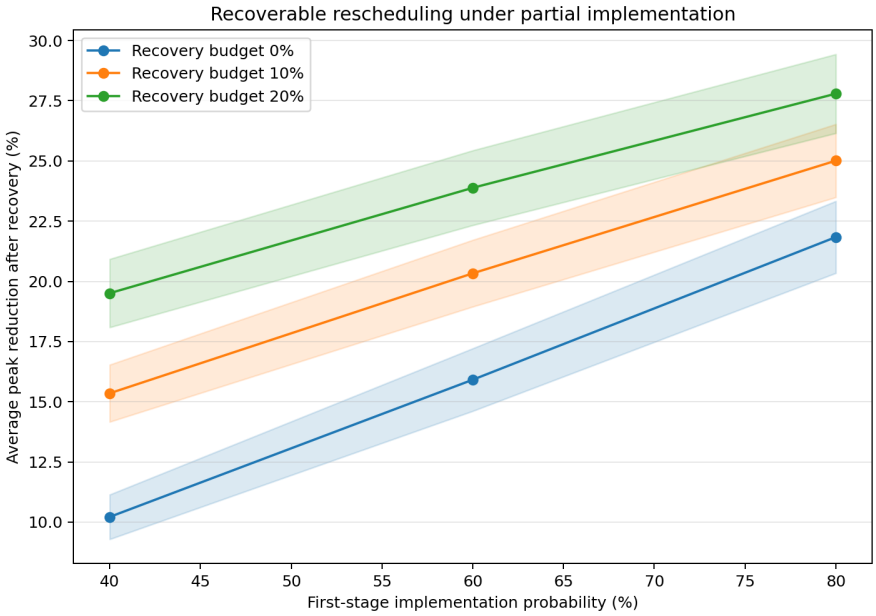


Figure 8: Average realized peak reduction with and without second-stage recovery, including 95% confidence bands

Figure 9 focuses on the incremental value of recovery. The largest incremental gains occur when the implementation probability is low or moderate. At 40% implementation probability, a 20% recovery budget adds 9.29 percentage points of peak reduction. At 60%, the corresponding gain is 7.97 percentage points. At 80%, the gain is smaller but still meaningful at 5.96 percentage points.

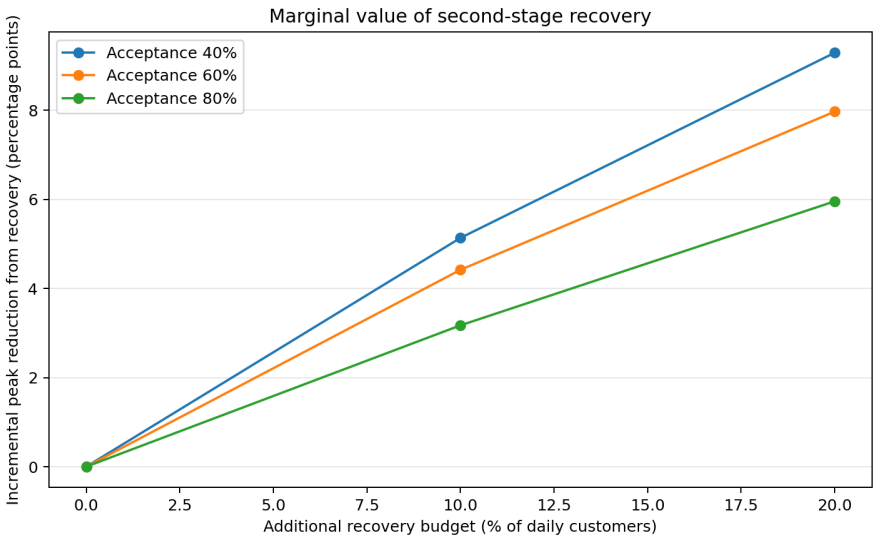


Figure 9: Incremental value of second-stage recovery

The recoverable rescheduling experiment adds three insights to the numerical study. First, partial implementation does not have to be treated as the final outcome of the planning process. A limited second-stage recovery action can restore a relevant share of the lost smoothing benefit. Second, the recovery value is obtained with relatively few additional changes. In the 60% implementation case, the 20% recovery budget uses 2.69 additional customer moves per day on average, yet increases realized peak reduction by 7.97 percentage points. Third, the same minimax profile-placement model remains useful after first-stage implementation information is known. The model can first be used to create the initial schedule adjustment and then be used again as a recovery tool.

Overall, the recovery extension strengthens the managerial interpretation of the model without broadening the thesis beyond its core. The deterministic results show the smoothing potential under controlled rescheduling. The recovery analysis then shows that part of this value can be restored if implementation is incomplete. This supports a layered decision-support view: optimize the initial proposal, observe which parts of the proposal can be implemented, and use limited recovery only when necessary.

5.5 Overall Numerical Findings

The numerical study produces five main findings. First, the original schedules contain a strong gap between average load and peak load. The average peak of 32.03 trucks is more than twice the average slot load of 14.69 trucks. This indicates that smoothing is a relevant planning objective because the required capacity level is driven by temporal overlap at peak times.

Second, customer flexibility has substantial value. With ± 15 minutes flexibility, the model reduces the peak by 11.88% on average. With ± 30 minutes, the reduction increases to 22.85%, and with ± 60 minutes it reaches 36.07%. These results suggest that even modest flexibility around the original schedule can materially reduce the highest capacity requirement.

Third, disruption budgets produce a useful decision frontier. A 20% movement budget captures 50.00% of the unrestricted peak-reduction benefit while moving only 17.49% of customers on average. A 50% movement budget captures 82.95% of the unrestricted benefit while moving 44.14% of customers. These numbers provide a concrete way to discuss rescheduling intensity. A planner can compare how much additional peak reduction is gained by increasing the allowed intervention in the schedule.

Fourth, the day-level analysis shows that the model improves every daily instance when movement is allowed, although the magnitude of improvement depends on the original overlap structure. This means that the model is not only an average-performance tool; it can also be used diagnostically to identify days on which rescheduling is especially valuable.

Fifth, recoverable rescheduling adds an operational response mechanism after partial first-stage implementation has been observed. At 60% implementation probability, the pre-recovery peak reduction is 15.91%. A 10% recovery budget increases it to 20.33%, and a 20% recovery budget increases it to 23.88%.

The larger recovery budget restores approximately 61.34% of the deterministic benefit lost through partial implementation while moving only 2.69 additional customers per day on average. This result shows that the same minimax model can be used not only to create the first proposal, but also to support a limited second-stage recovery action.

Overall, the numerical results support the proposed modelling approach. The minimax rescheduling model identifies high-impact customer profile movements, quantifies the value of additional flexibility, and provides a transparent efficiency-disruption frontier. The most relevant operating range appears to lie between the 20% and 50% movement-budget cases. The 20% budget is attractive because it obtains half of the unrestricted benefit with a relatively small intervention. The 50% budget is attractive because it obtains more than four fifths of the unrestricted benefit while still avoiding the broader disruption of the unrestricted solution. The recovery analysis refines this picture by showing that limited second-stage adjustment can restore a meaningful share of the benefit if part of the first-stage plan cannot be implemented. These results form the quantitative basis for the managerial insights developed in the following chapter.

6 Managerial Insights

This chapter translates the numerical results into managerial implications. The main message is that the minimax rescheduling model should not be interpreted as a fully automated dispatching system. Its practical value lies in providing a transparent decision-support layer for diagnosing peak pressure, selecting high-impact customer movements, and controlling the amount of disruption created by rescheduling.

6.1 Peak Diagnosis and Rescheduling Intensity

The first managerial use of the model is diagnostic. The original schedules have an average peak of 32.03 trucks, while the average slot load is only 14.69 trucks. This means that the capacity problem is not simply caused by high total demand. It is caused by the timing of overlapping customer profiles. A planner who only observes the daily total volume would miss this structure. The model makes the structure visible by translating every customer profile into a contribution to the aggregate load curve.

This diagnostic role is important because it shifts the planning discussion from general capacity pressure to specific peak-causing profiles. Instead of asking whether the entire day is overloaded, the planner can ask which customer profiles create the highest marginal contribution around the peak. The optimization output then identifies a small set of movements that creates a large part of the available smoothing effect.

The disruption-budget results suggest that the most attractive operating range lies between moderate and substantial, but not unrestricted, rescheduling. A 20% movement budget captures 50.00% of the unrestricted peak-reduction benefit while moving only 17.49% of customers on average. A 50% movement budget captures 82.95% of the unrestricted benefit while moving 44.14% of customers. In contrast, the unrestricted case moves 74.27% of customers on average.

This pattern is managerially important because it shows diminishing returns. The first set of movements is highly valuable; later movements still improve the peak but create less incremental benefit per additional customer touched. Therefore, the model should not be used only in its unrestricted form. A disruption budget is not a technical constraint added for convenience. It is a managerial control parameter that allows the planner to choose how aggressive the

rescheduling intervention should be.

6.2 Capacity-Tier Decisions and Service Policy

Peak smoothing matters because capacity is usually organized in tiers. A firm may have an owned fleet or stable internal capacity, then predictable long-term rented capacity, and finally more expensive or less reliable short-term capacity. The minimax objective directly targets the highest simultaneous requirement and is therefore useful when the managerial question is how much capacity must be available at the busiest moment of the day.

The model can support tactical capacity discussions even before a full routing model is introduced. For example, if the baseline peak regularly exceeds a preferred internal capacity level, the planner can evaluate whether limited rescheduling brings the peak below that level. If it does, the firm may reduce reliance on expensive peak capacity. If it does not, the result still provides useful evidence that the peak is structurally difficult to smooth without broader flexibility or additional resources.

The flexibility sensitivity shows that time-window flexibility is an operational resource. Even ± 15 minutes reduces the average peak by 11.88%, while ± 60 minutes reduces it by 36.07%. This does not mean that every customer should automatically be moved by one hour. It means that the planner should define service policies that create controlled flexibility around the original schedule.

A practical policy could classify customers by permissible shift ranges. Some customers might allow only small adjustments, while others might allow broader windows due to receiving-hour flexibility or lower time criticality. The model can then use these customer-specific feasible sets. This would preserve the logic of the thesis while making the implementation more realistic. The important managerial insight is that flexibility should be explicitly managed rather than

treated as an informal exception.

6.3 Recovery and Data Requirements

The recovery experiment shows that the model can also support second-stage correction. If part of the first-stage plan cannot be implemented, the planner does not need to accept the resulting peak as final. A limited recovery budget can restore a meaningful share of the lost smoothing benefit. At 60% implementation probability, a 20% recovery budget increases peak reduction from 15.91% to 23.88% and restores 61.34% of the lost benefit.

This does not imply that recovery should be used as a routine substitute for careful first-stage planning. Recovery creates additional operational work and should therefore be treated as an exception mechanism. Its value is highest when the realized first-stage outcome is worse than expected or when the remaining peak would create a relevant capacity problem. In that sense, the recoverable model provides a structured escalation step: first optimize the initial proposal, then recover only if the implementation outcome makes recovery worthwhile.

A practical implementation would require several additions to the current data structure. The current model already needs customer-level demand profiles, original start times, profile durations, and feasible shift ranges. For operational use, the firm would also need customer-specific feasibility rules, service-level restrictions, capacity thresholds, and calibrated information about the cost of peak capacity. If the model is later connected to routing, it would additionally require customer locations, travel times, depot information, driver constraints, vehicle availability, and route feasibility checks.

The managerial value of the current model is therefore not that it solves the complete delivery-planning problem. Its value is that it isolates one operational

question clearly: how much peak-capacity pressure can be reduced by shifting complete customer demand profiles within limited flexibility windows? This layer can be implemented before, beside, or above more detailed routing tools.

7 Conclusion

This thesis developed and evaluated a minimax rescheduling model for smoothing time-dependent truck demand in B2B delivery scheduling. The starting point was an existing customer schedule with fixed demand profiles over a discrete daily timeline. The model chooses exactly one feasible final position for each customer profile and minimizes the maximum aggregate truck demand over all time slots. Disruption is controlled through movement budgets and lexicographic secondary objectives.

The main research question asked how customer delivery schedules can be rescheduled to minimize peak truck demand under limited customer flexibility and minimal schedule disruption. The numerical results answer this question in several steps. First, the model identifies customer profiles whose movements reduce costly overlap at the daily peak. Second, additional time-window flexibility increases smoothing potential: unrestricted ± 60 minute flexibility reduces the average peak by 36.07%. Third, disruption budgets reveal a clear efficiency-disruption frontier. A 20% movement budget captures half of the unrestricted benefit, and a 50% movement budget captures more than four fifths. Fourth, the day-level analysis shows that improvements occur across all daily instances, not only in a few exceptional cases. Fifth, recoverable rescheduling shows that limited second-stage correction can restore a meaningful share of the benefit if part of the first-stage plan cannot be implemented.

The contribution of the thesis is both methodological and managerial. Methodologically, the thesis formulates a tractable profile-placement MILP for tem-

poral capacity smoothing and links it with disruption control and recoverable rescheduling. Managerially, it shows that peak truck demand can be reduced without changing total demand or solving a full routing problem. The model provides a transparent decision-support structure for deciding which customers to contact, how much rescheduling effort is likely to be worthwhile, and when a limited recovery step may be useful.

The results are subject to limitations. The model abstracts from locations, travel times, depots, individual vehicles, drivers, and route sequences. It therefore does not prove that every optimized temporal schedule would be directly executable in a full dispatching system. The generated instance set is realistic enough for testing the optimization logic, but future empirical validation with company data would be valuable. The capacity-tier interpretation is also a planning proxy rather than a fully calibrated monetary business case, because real cost coefficients for owned, rented, and spot-market capacity are not included. These limitations delimit the interpretation of the results, but they do not undermine the central finding that temporal profile placement is a useful analytical layer.

Future research could extend the model in several directions. The most immediate extension would be to combine the profile-placement model with routing feasibility checks, so that temporal smoothing does not create infeasible routes. A second direction would calibrate capacity-tier costs and translate peak-load reductions into expected cost savings. A third extension would use historical operational data to estimate customer-specific feasible shift ranges and recurring demand-profile patterns. Finally, larger empirical datasets could test whether the diminishing-return pattern observed in the numerical study also appears in real B2B delivery systems.

Overall, the thesis shows that temporal rescheduling can be a powerful and interpretable lever for reducing peak simultaneous truck demand. The deter-

ministic minimax model forms the core of the approach, while the recoverable extension demonstrates that the same modelling logic can also support limited corrective action when implementation conditions change.

Bibliography

Abdollahi, M., Yang, X., Nasri, M. I., and Fairbank, M. (2023). Demand management in time-slotted last-mile delivery via dynamic routing with forecast orders. *European Journal of Operational Research*, 309(2), 704–718. https://doi.org/10.1016/j.ejor.2023.01.023

Agatz, N., Fleischmann, M., and van Nunen, J. A. E. E. (2008). E-fulfillment and multi-channel distribution: A review. *European Journal of Operational Research*, 187(2), 339–356. https://doi.org/10.1016/j.ejor.2007.04.024

Agatz, N., Campbell, A. M., Fleischmann, M., and Savelsbergh, M. W. P. (2011). Time slot management in attended home delivery. *Transportation Science*, 45(3), 435–449. https://doi.org/10.1287/trsc.1100.0346

Asbach, L., Dorndorf, U., and Pesch, E. (2009). Analysis, modeling and solution of the concrete delivery problem. *European Journal of Operational Research*, 193(3), 820–835. https://doi.org/10.1016/j.ejor.2007.11.011

Asdemir, K., Jacob, V. S., and Krishnan, R. (2009). Dynamic pricing of multiple home delivery options. *European Journal of Operational Research*, 196(1), 246–257. https://doi.org/10.1016/j.ejor.2008.03.005

Aytug, H., Lawley, M. A., McKay, K., Mohan, S., and Uzsoy, R. (2005). Executing production schedules in the face of uncertainties: A review and some future directions. *European Journal of Operational Research*, 161(1), 86–110. https://doi.org/10.1016/j.ejor.2003.08.027

Bent, R. and Van Hentenryck, P. (2004). Scenario-based planning for partially dynamic vehicle routing with stochastic customers. *Operations Research*, 52(6), 977–987. https://doi.org/10.1287/opre.1040.0124

Bräysy, O. and Gendreau, M. (2005a). Vehicle routing problem with time win-

dows, Part I: Route construction and local search algorithms. *Transportation Science*, 39(1), 104–118. https://doi.org/10.1287/trsc.1030.0056

Bräysy, O. and Gendreau, M. (2005b). Vehicle routing problem with time windows, Part II: Metaheuristics. *Transportation Science*, 39(1), 119–139. https://doi.org/10.1287/trsc.1030.0057

Campbell, A. M. and Savelsbergh, M. W. P. (2005). Decision support for consumer direct grocery initiatives. *Transportation Science*, 39(3), 313–327. https://doi.org/10.1287/trsc.1040.0105

Campbell, A. M. and Savelsbergh, M. W. P. (2006). Incentive schemes for attended home delivery services. *Transportation Science*, 40(3), 327–341. https://doi.org/10.1287/trsc.1050.0136

Daganzo, C. F. (1987). Modeling distribution problems with time windows. Part I. *Transportation Science*, 21(3), 171–179. https://doi.org/10.1287/trsc.21.3.171

Daniels, R. L. and Kouvelis, P. (1995). Robust scheduling to hedge against processing time uncertainty in single-stage production. *Management Science*, 41(2), 363–376. https://doi.org/10.1287/mnsc.41.2.363

Dror, M., Laporte, G., and Trudeau, P. (1989). Vehicle routing with stochastic demands: Properties and solution frameworks. *Transportation Science*, 23(3), 166–176. https://doi.org/10.1287/trsc.23.3.166

Durbin, M. and Hoffman, K. (2008). OR Practice—The dance of the thirty-ton trucks: Dispatching and scheduling in a dynamic environment. *Operations Research*, 56(1), 3–19. https://doi.org/10.1287/opre.1070.0459

Ehmke, J. F. and Campbell, A. M. (2014). Customer acceptance mechanisms

for home deliveries in metropolitan areas. *European Journal of Operational Research*, 233(1), 193–207. https://doi.org/10.1016/j.ejor.2013.08.028

Feng, C.-W., Cheng, T.-M., and Wu, H.-T. (2004). Optimizing the schedule of dispatching RMC trucks through genetic algorithms. *Automation in Construction*, 13(3), 327–340. https://doi.org/10.1016/j.autcon.2003.10.001

Graham, R. L., Lawler, E. L., Lenstra, J. K., and Rinnooy Kan, A. H. G. (1979). Optimization and approximation in deterministic sequencing and scheduling: A survey. *Annals of Discrete Mathematics*, 5, 287–326. https://doi.org/10.1016/S0167-5060(08)70356-X

Herroelen, W. and Leus, R. (2005). Project scheduling under uncertainty: Survey and research potentials. *European Journal of Operational Research*, 165(2), 289–306. https://doi.org/10.1016/j.ejor.2004.04.002

Hvattum, L. M., Løkketangen, A., and Laporte, G. (2006). Solving a dynamic and stochastic vehicle routing problem with a sample scenario hedging heuristic. *Transportation Science*, 40(4), 421–438. https://doi.org/10.1287/trsc.1060.0166

Kinable, J., Wauters, T., and Vanden Berghe, G. (2014). The concrete delivery problem. *Computers & Operations Research*, 48, 53–68. https://doi.org/10.1016/j.cor.2014.02.008

Klein, R. S., Luss, H., and Rothblum, U. G. (1993). Minimax resource allocation problems with resource-substitutions represented by graphs. *Operations Research*, 41(5), 959–971. https://doi.org/10.1287/opre.41.5.959

Klein, R., Neugebauer, M., Ratkovitch, D., and Steinhardt, C. (2019). Differentiated time slot pricing under routing considerations in attended home delivery. *Transportation Science*, 53(1), 236–255. https://doi.org/10.1287/trsc.2017.0738

Koch, S. and Klein, R. (2020). Route-based approximate dynamic programming for dynamic pricing in attended home delivery. *European Journal of Operational Research*, 287(2), 633–652. https://doi.org/10.1016/j.ejor.2020.04.002

Köhler, C., Ehmke, J. F., and Campbell, A. M. (2020). Flexible time window management for attended home deliveries. *Omega*, 91, 102023. https://doi.org/10.1016/j.omega.2019.01.001

Kouvelis, P. and Yu, G. (1997). *Robust Discrete Optimization and Its Applications*. Kluwer Academic Publishers, Dordrecht. https://doi.org/10.1007/978-1-4757-2620-6

Kuster, J., Jannach, D., and Friedrich, G. (2010). Applying local rescheduling in response to schedule disruptions. *Annals of Operations Research*, 180, 265–282. https://doi.org/10.1007/s10479-008-0488-x

Lin, P.-C., Wang, J., Huang, S.-H., and Wang, Y.-T. (2010). Dispatching ready mixed concrete trucks under demand postponement and weight limit regulation. *Automation in Construction*, 19(6), 798–807. https://doi.org/10.1016/j.autcon.2010.05.002

Matsatsinis, N. F. (2004). Towards a decision support system for the ready concrete distribution system: A case of a Greek company. *European Journal of Operational Research*, 152(2), 487–499. https://doi.org/10.1016/S0377-2217(03)00039-0

Naso, D., Surico, M., Turchiano, B., and Kaymak, U. (2007). Genetic algorithms for supply-chain scheduling: A case study in the distribution of ready-mixed concrete. *European Journal of Operational Research*, 177(3), 2069–2099. https://doi.org/10.1016/j.ejor.2005.12.019

Ouelhadj, D. and Petrovic, S. (2009). A survey of dynamic schedul-

ing in manufacturing systems. *Journal of Scheduling*, 12(4), 417–431. https://doi.org/10.1007/s10951-008-0090-8

Pillac, V., Gendreau, M., Guéret, C., and Medaglia, A. L. (2013). A review of dynamic vehicle routing problems. *European Journal of Operational Research*, 225(1), 1–11. https://doi.org/10.1016/j.ejor.2012.08.015

Pinedo, M. (1983). Stochastic scheduling with release dates and due dates. *Operations Research*, 31(3), 559–572. https://doi.org/10.1287/opre.31.3.559

Rangsaritratsamee, R., Ferrell, W. G., and Kurz, M. B. (2004). Dynamic rescheduling that simultaneously considers efficiency and stability. *Computers & Industrial Engineering*, 46(1), 1–15. https://doi.org/10.1016/j.cie.2003.09.007

Rieck, J., Zimmermann, J., and Gather, T. (2012). Mixed-integer linear programming for resource leveling problems. *European Journal of Operational Research*, 221(1), 27–37. https://doi.org/10.1016/j.ejor.2012.03.003

Savelsbergh, M. W. P. (1985). Local search in routing problems with time windows. *Annals of Operations Research*, 4(1), 285–305. https://doi.org/10.1007/BF02022044

Schmid, V., Doerner, K. F., Hartl, R. F., and Salazar-González, J.-J. (2010). Hybridization of very large neighborhood search for ready-mixed concrete delivery problems. *Computers & Operations Research*, 37(3), 559–574. https://doi.org/10.1016/j.cor.2008.07.010

Solomon, M. M. (1987). Algorithms for the vehicle routing and scheduling problems with time window constraints. *Operations Research*, 35(2), 254–265. https://doi.org/10.1287/opre.35.2.254

Strauss, A., Gülpınar, N., and Zheng, Y. (2021). Dynamic pricing of flexible time

slots for attended home delivery. *European Journal of Operational Research*, 294(3), 1022–1041. https://doi.org/10.1016/j.ejor.2020.03.007

Suresh, V. and Chaudhuri, D. (1993). Dynamic scheduling - A survey of research. *International Journal of Production Economics*, 32(1), 53–63. https://doi.org/10.1016/0925-5273(93)90007-8

Tighazoui, A., Sauvey, C., and Sauer, N. (2021). Predictive-reactive strategy for identical parallel machine rescheduling. *Computers & Operations Research*, 134, 105372. https://doi.org/10.1016/j.cor.2021.105372

Ulmer, M. W. (2020). Dynamic pricing and routing for same-day delivery. *Transportation Science*, 54(4), 1016–1033. https://doi.org/10.1287/trsc.2019.0958

Ulmer, M. W., Söffker, N., and Mattfeld, D. C. (2018). Value function approximation for dynamic multi-period vehicle routing. *European Journal of Operational Research*, 269(3), 883–899. https://doi.org/10.1016/j.ejor.2018.02.038

van den Akker, M., Hoogeveen, H., and Stoef, J. (2018). Combining two-stage stochastic programming and recoverable robustness to minimize the number of late jobs in the case of uncertain processing times. *Journal of Scheduling*, 21, 607–617. https://doi.org/10.1007/s10951-018-0559-z

Vieira, G. E., Herrmann, J. W., and Lin, E. (2000). Analytical models to predict the performance of a single-machine system under periodic and event-driven rescheduling strategies. *International Journal of Production Research*, 38(8), 1899–1915. https://doi.org/10.1080/002075400188654

Vieira, G. E., Herrmann, J. W., and Lin, E. (2003). Rescheduling manufacturing systems: A framework of strategies, policies, and methods. *Journal of Scheduling*, 6, 39–62. https://doi.org/10.1023/A:1022235519958

Wu, S. D., Storer, R. H., and Chang, P.-C. (1993). One-machine rescheduling

heuristics with efficiency and stability as criteria. *Computers & Operations Research*, 20(1), 1–14. https://doi.org/10.1016/0305-0548(93)90091-V

Yan, S., Lai, W., and Chen, M. (2008). Production scheduling and truck dispatching of ready mixed concrete. *Transportation Research Part E: Logistics and Transportation Review*, 44(1), 164–179. https://doi.org/10.1016/j.tre.2006.05.001

Yang, X. and Strauss, A. K. (2017). An approximate dynamic programming approach to attended home delivery management. *European Journal of Operational Research*, 263(3), 935–945. https://doi.org/10.1016/j.ejor.2017.06.034

Yang, X., Strauss, A. K., Currie, C. S. M., and Eglese, R. (2016). Choice-based demand management and vehicle routing in e-fulfillment. *Transportation Science*, 50(2), 473–488. https://doi.org/10.1287/trsc.2014.0549

List of Tools and Aids Used

Academic sources. All academic publications and other external sources used in the thesis are identified in the bibliography and cited at the relevant points in the text.

Data and computational environment. The numerical study uses an empirically motivated generated dataset of customer-level delivery requests. The model implementation and analysis were carried out in Python using NumPy, pandas, SciPy (including `scipy.optimize.milp` and sparse-matrix utilities), Matplotlib, and Jupyter. The mixed-integer programs were solved through the solver interface provided by SciPy. The thesis was typeset and assembled in LaTeX.

Generative-AI tools. OpenAI ChatGPT/Codex was used as a supporting aid for drafting, literature-summary support, idea and outline structuring, grammar and language revision, LaTeX formatting, consistency checks, and review/debugging support for parts of the Python implementation. All cited sources, mathematical formulations, code, numerical results, interpretations, and final wording were independently reviewed by the author. Generative-AI output was not treated as an academic source.

Responsibility. The research question, scope, modelling choices, empirical interpretation, and final decisions are the author's own. The author assumes full responsibility for the accuracy and integrity of the submitted work.

Progress Documentation

Student: Murat Tutar (matriculation number 7431010)
Thesis: *A Minimax Rescheduling Model for Smoothing Time-Dependent Truck Demand in B2B Delivery Scheduling*

Development of the research focus and incorporation of feedback

The thesis developed iteratively from a broad delivery-planning idea into a focused optimization study. At the outset, I examined what could be modelled rigorously with the available customer-level data. Because the data contained start times and time-dependent truck-demand profiles but no customer locations, routes, depots, driver rosters, or vehicle assignments, I defined the project as a temporal profile-rescheduling problem rather than a vehicle-routing problem. This boundary became the basis for the research question and for the structure of the thesis.

In the modelling phase, the central objective was specified as a minimax problem: the schedule should reduce the largest simultaneous truck requirement while keeping changes operationally acceptable. I incorporated this direction by formulating a deterministic profile-placement model with limited shift windows, changed-customer budgets, and lexicographic disruption control. I also added a small illustrative example before the formal model so that the profile-placement logic and the peak-reduction mechanism are transparent.

The mandatory consultation took place on 7 July 2026. The discussion covered the status of the work, the topic and research question, the objective, the methodology, the opportunities and limitations of generative AI, and the importance of scientific rigour. The signed consultation report records two central pieces of advice: maintain consistency between the structure and the research question, and keep the thesis focused on that question. I used this feedback as a final test for the outline. The literature review was reorganized around rescheduling and stability, resource smoothing and minimax allocation, timed delivery operations, and recoverable rescheduling. Broader customer-acceptance and stochastic extensions were removed from the final thesis. Recoverable rescheduling was retained only as a limited second-stage extension of the same deterministic core.

The numerical chapter was then aligned with the revised outline. It first establishes the baseline load structure, then evaluates flexibility and movement budgets, and finally tests the focused recovery extension. The managerial-insights chapter was revised to follow directly from those results, particularly the implications for peak diagnosis, movement-budget selection, capacity tiers, service policy, and recovery planning. This made the research question the organizing thread from the literature review through the model, experiments, and conclusion.

Milestones, learning progress, and reflection on working methods

The main milestones were: (1) delimiting the planning problem and evaluating the available data; (2) defining the deterministic minimax model and disruption measures; (3) implementing the model and validating it on generated planning instances; (4) conducting flexibility, movement-budget, and day-level experiments; (5) completing the midterm presentation and consultation; and (6) narrowing and restructuring the final thesis in response to the feedback. Working through these stages showed me that methodological focus is not achieved by adding every feasible extension. It is achieved by selecting the analyses that answer the research question most clearly and by making the limits of the data explicit.

My modelling practice also became more disciplined during the project. I learned to separate the core objective from secondary criteria, to test whether constraints matched their operational interpretation, and to evaluate solution quality together with schedule disruption. The generated data were suitable for controlled model testing, but they do not justify claims about routing feasibility or realized cost savings. Stating these boundaries explicitly improved both the scientific precision and the managerial interpretation of the work.

Use of generative AI

Generative AI was used as a supporting tool during the thesis process. As recorded in the consultation report, it was discussed and used especially for literature-summary support and grammar correction. It also supported drafting, idea and outline structuring, language polishing, LaTeX formatting, consistency checks, and the review/debugging of parts of the Python implementation.

The substantive responsibility remained with me throughout. I selected the research focus, evaluated and incorporated supervisory feedback, chose the modelling assumptions, checked the implementation and numerical results, verified the cited sources, interpreted the findings, and decided which material to retain. The final thesis and its accompanying documentation were reviewed by me, and I assume full responsibility for them.

UNIVERSITÄT ZU KÖLN

DEPARTMENT OF OPERATIONS MANAGEMENT

Consultation Report on Progress and Key Findings from the Thesis

Name, surname:

Murat Tutar

Matriculation number:

7431010

Degree program:

Business Administration, Schwerpunkt/Major SCM

Title of thesis:

A Minimax Rescheduling Model for Smoothing Time-Dependent Truck Demand in B2B Delivery Scheduling

Information about the Counseling Session

Initial reviewer:

Jun.-Prof. Dr. Florian E. Sachs

Second reviewer:

Prof. Ulrich W. Thonemann, Ph.D

If necessary, another related/competent person:

Date of counseling session:

07.07.2026

Time of counseling session (from/to):

09:00 - 12:00

Protocol Sheet

Student's Expectations for the Conversation

At the beginning of the conversation, the student expressed the following wishes or expectations:

- [ ] Gaining confidence
- [x] Receive suggestions
- [x] Answering open questions

Further:

Content of the Conversation

- [x] Status of work
- [ ] Workflow
- [x] Topic and research question
- [x] Objective of the work
- [ ] Theoretical framework
- [x] Methodology
- [ ] Arguments and findings
- [ ] Sources
- [ ] Organizational matters

Further:

Use of AI in the Editing Process

- [x] Opportunities offered by AI
- [x] Limitations of AI
- [x] Influence of AI on the editing process
- [x] The importance of scientific rigour with AI

Further:

Discussed using AI, esp. for literature summary and grammar correction.

Further advice from the advisor to the student

- [x] Consistency between structure and research question
- [x] Focusing on the research question
- [ ] Alignment of arguments and findings with the research question, theory, and methodology
- [ ] Consistency, coherence, and stringency of argumentation

Further:

Follow-up appointment

A follow-up appointment is planned. The specific date will be scheduled by request.

Additional notes

07.07.2026 Digital unterschrieben von Florian Elias Sachs (Jun.-Prof. Dr. Florian E. Sachs)

Signature